□□□□

□1□ □□□□

□□□□□□ near-zero mode □□□□□□□

□2□ □□□□

old_mode: □□□□□□□□ + □□□□□□
best_edge_state: □□□□□□□□ W_edge_total □□□□□
best_corner_state: □□□□□□□□ W_corner □□□□□
optional_best_side_state: left/right/top/bottom □□□□□

□□□□□□□ delta_E □□□□□□□□□□□ P_edge □ P_corner □□□□□□
□□ maximally edge/corner-localized rotated state□

□□□□□□□□□□

```
{
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    "nx": 20,
    "ny": 20,
    "base_k": 192,
    "min_candidate_states": 16
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  "energy_windows": {
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    "zero_half_width": 0.25,
    "ldos_half_width": 0.05,
    "fallback_half_width": 0.08
  },
  "geometry": {
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    "corner_size": 3
  },
  "degeneracy": {
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    "relative_delta_e": 0.01
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    "edge_threshold": 0.45
  },
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    "min_keep": 2,
    "max_keep": 8
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}
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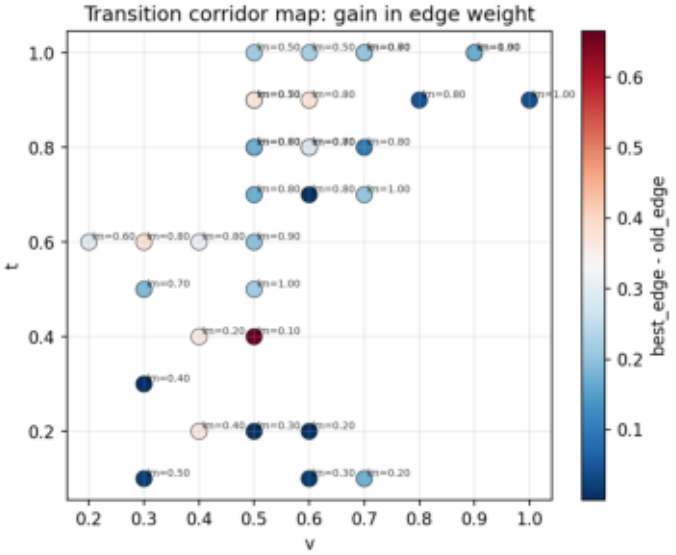
□3□ □□□□

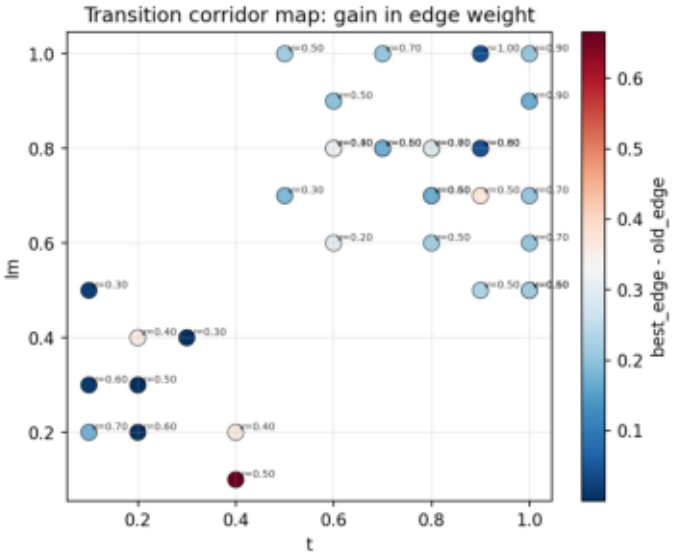
- □□□□□ 34/34 □□□□□□□□□□□
- □□□□□□□□□□best_edge-old□: 0.1948
- □□□□□□□□□□best_corner-old□: 0.1327

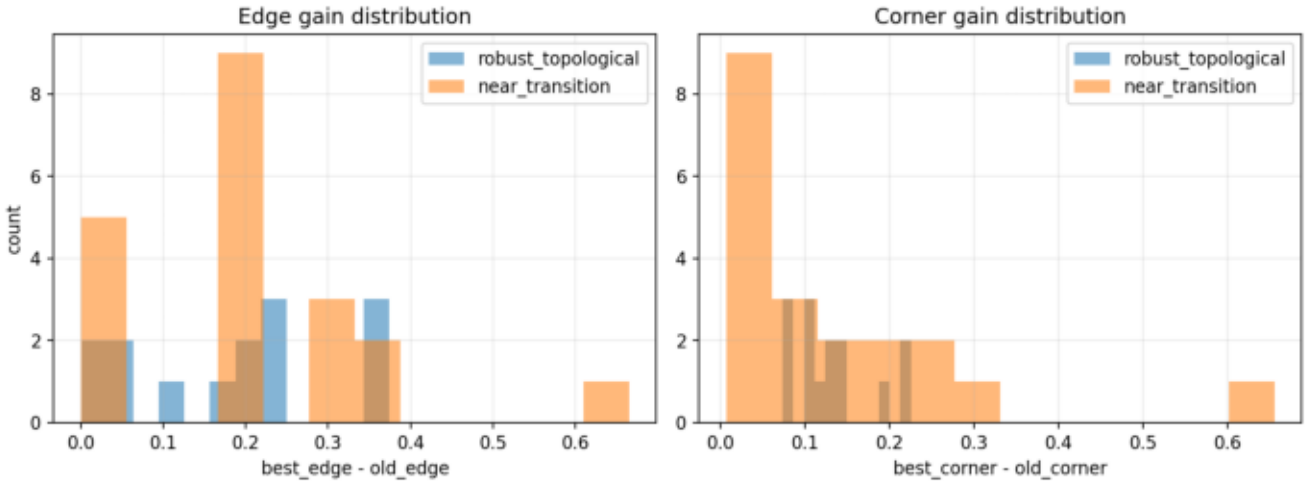
- robust_topological □□: 14
- near_transition □□: 20

□□□□□□□□
1) □□□□□□□□□□□“□□□□□□□□□”□
2) □□□□□□□□□ bulk/mixed □□□□□
3) □□“□□□□ + □□□□□□”□□□□□

□4□ □□□□□□□□□□□□


Transition corridor map: gain in edge weight
t
v
best_edge - old_edge


Transition corridor map: gain in edge weight
lm
t
best_edge - old_edge


Edge gain distribution
robust_topological
near_transition
count
best_edge - old_edge
Corner gain distribution
robust_topological
near_transition
best_corner - old_corner

□5□ □□□□□□□□□

□□□□□□□□□□□□best_edge-old > 0.08□: 25
□□□□□□ bulk/mixed □□□□: 1

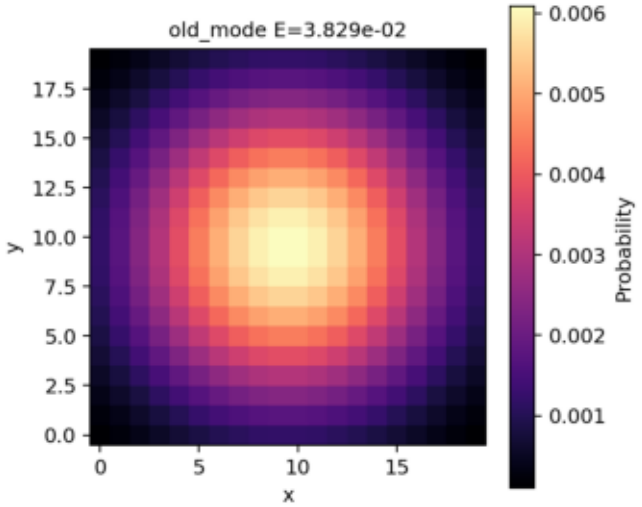
□□□□□□□□□□□□

□6□ □□□□□

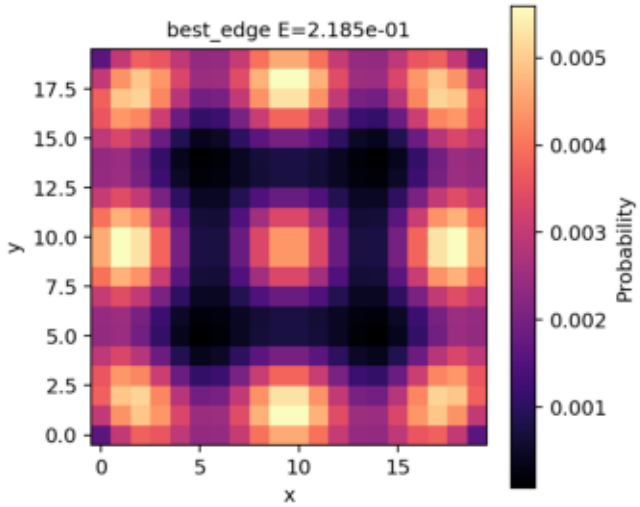
1) □□□□□□□□□20x20 flake □□□□□□□□□
2) ribbon □ flake □□□□□□□□□□□□□□□
3) Z2 □□□□□□□□□□□□
4) □□□□□□□□ LDOS □□□□□□□□□□□□

□□A - v_0p60_t_0p90_lm_0p80
v=0.60, t=0.90, lm=0.80, gap=2.000e-01, Z2=1, Chern=0.000, Wilson=2.626

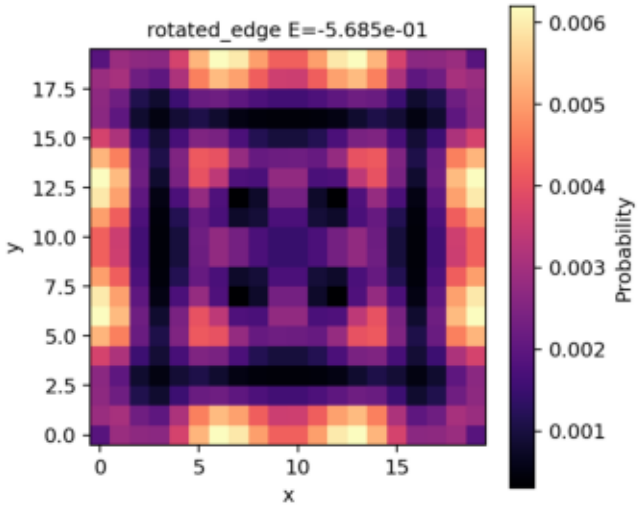
old



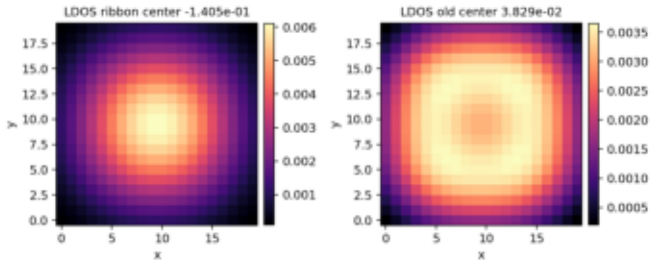
best_edge



rot_edge



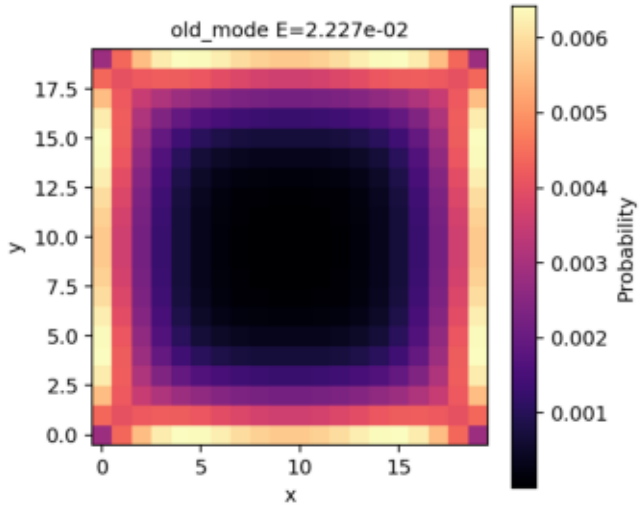
LDOS



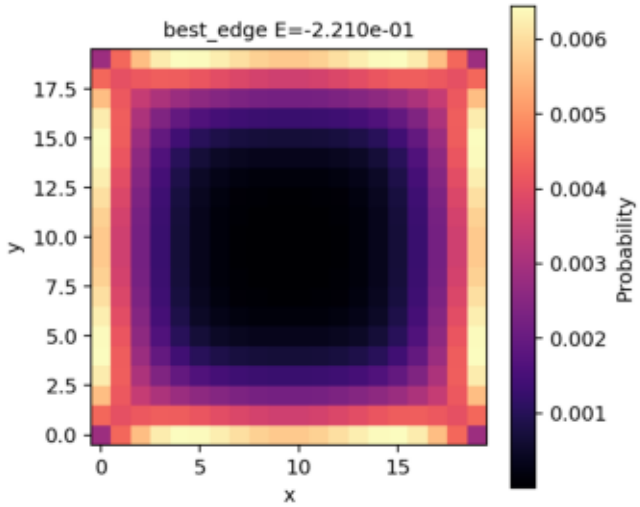
old_edge=0.139, best_edge=0.512, rot_edge=0.570; role=edge_dominated

□□A - v_0p60_t_0p80_lm_0p70
v=0.60, t=0.80, lm=0.70, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=0.661

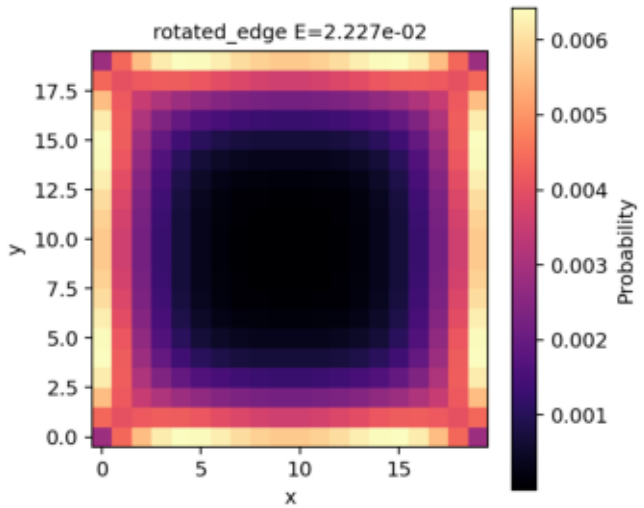
old



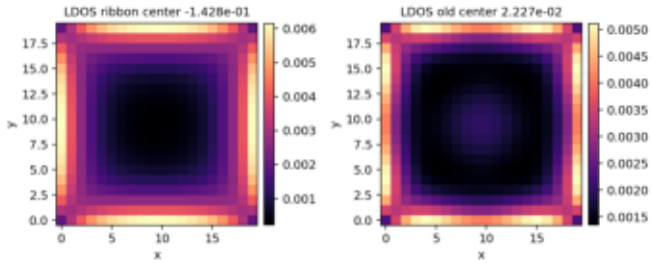
best_edge



rot_edge



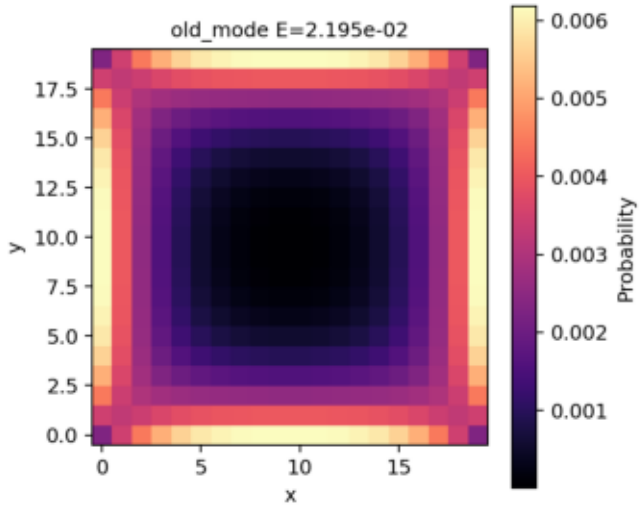
LDOS



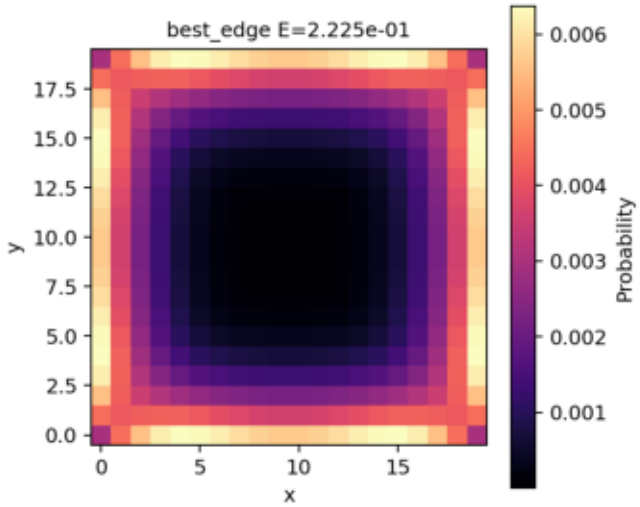
old_edge=0.703, best_edge=0.705, rot_edge=0.703; role=edge_dominated

□□A - v_1p00_t_0p90_lm_1p00
v=1.00, t=0.90, lm=1.00, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=0.823

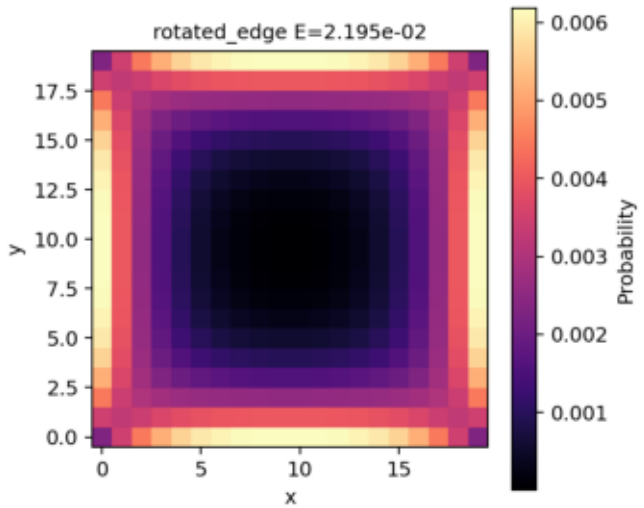
old



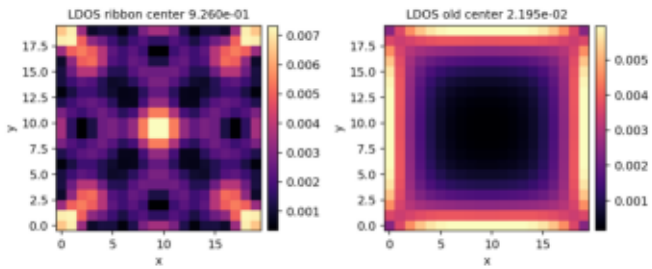
best_edge



rot_edge



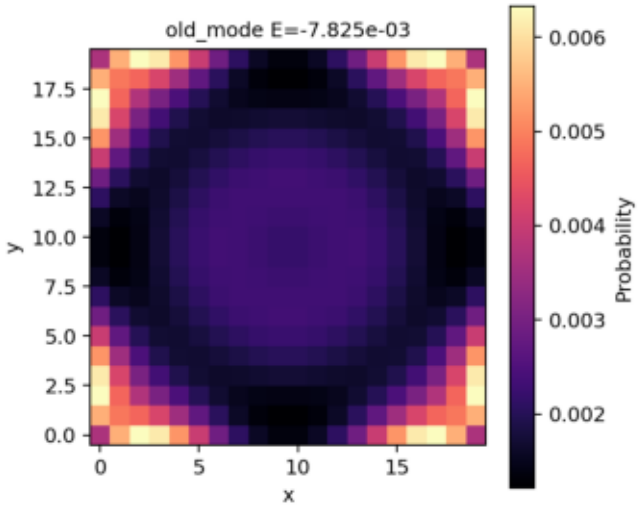
LDOS



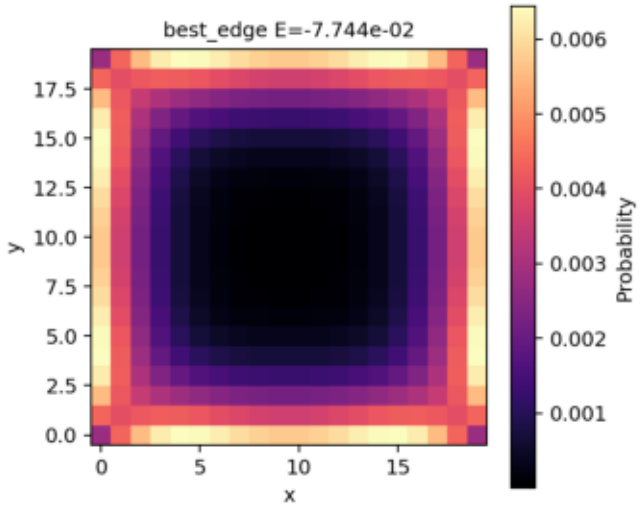
old_edge=0.661, best_edge=0.702, rot_edge=0.661; role=edge_dominated

□□A - v_0p50_t_0p80_lm_0p60
v=0.50, t=0.80, lm=0.60, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=-0.538

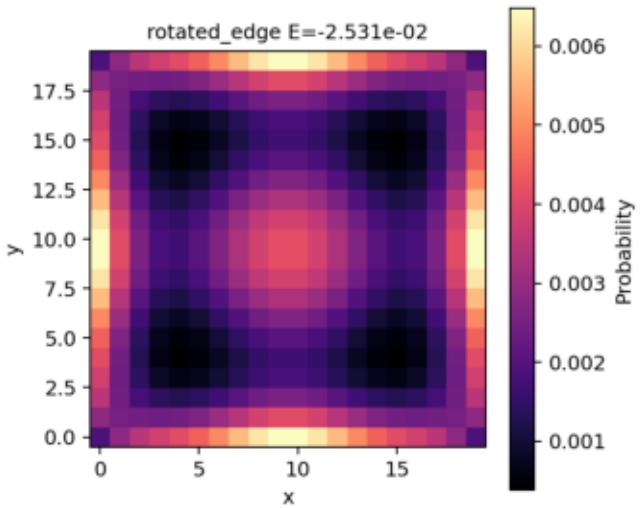
old



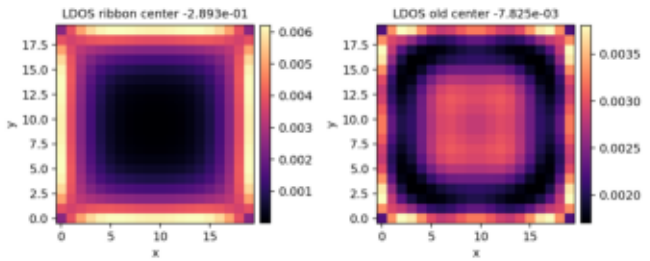
best_edge



rot_edge



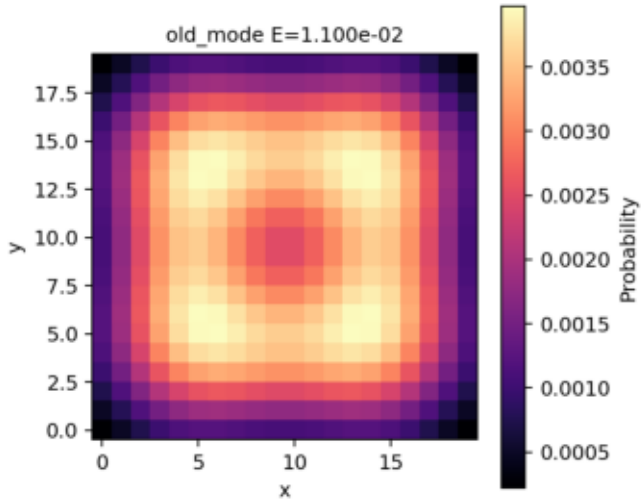
LDOS



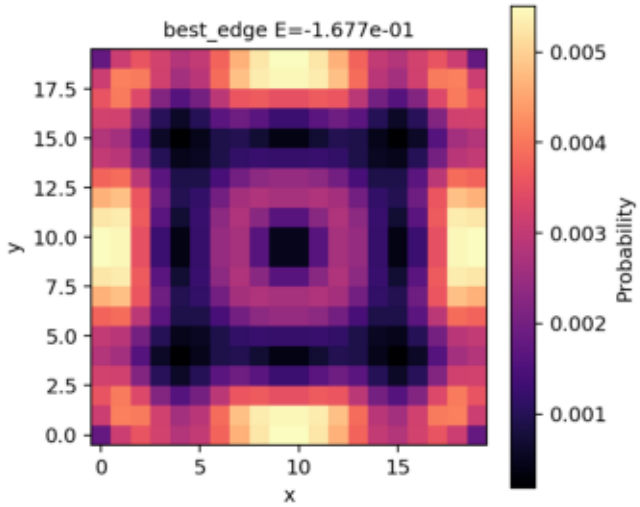
old_edge=0.484, best_edge=0.704, rot_edge=0.547; role=edge_dominated

□□A - v_0p30_t_0p60_lm_0p80
v=0.30, t=0.60, lm=0.80, gap=2.000e-01, Z2=1, Chern=0.000, Wilson=0.575

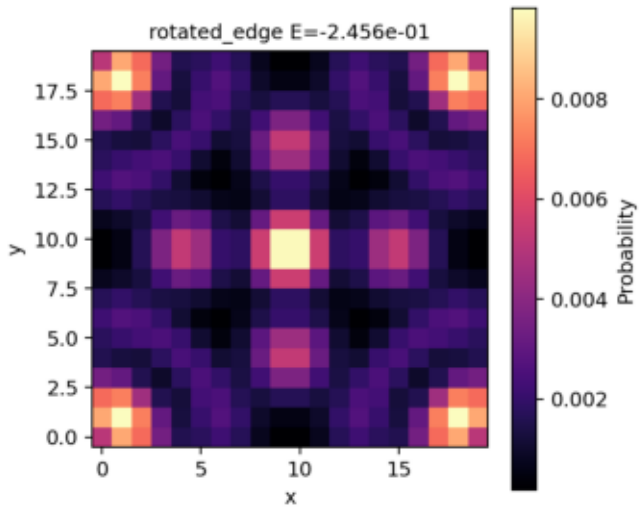
old



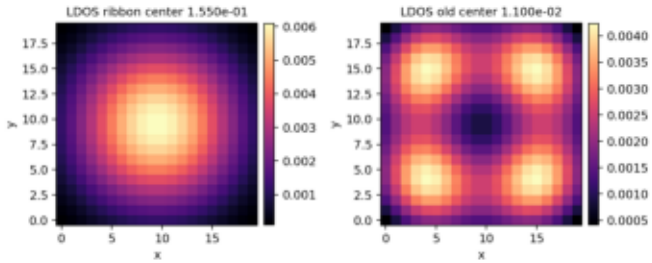
best_edge



rot_edge



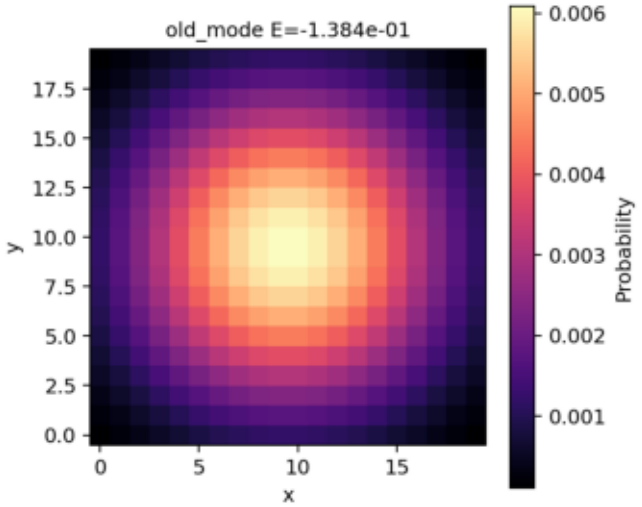
LDOS



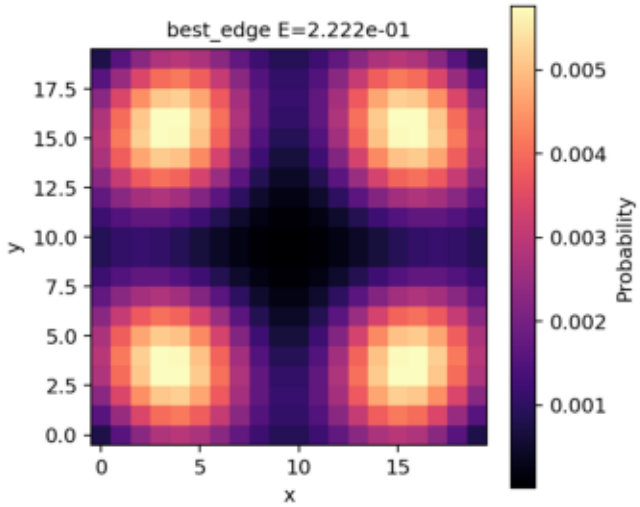
old_edge=0.186, best_edge=0.560, rot_edge=0.430; role=bulk_mixed

□□A - v_0p90_t_1p00_lm_1p00
v=0.90, t=1.00, lm=1.00, gap=2.000e-01, Z2=1, Chern=0.000, Wilson=-0.605

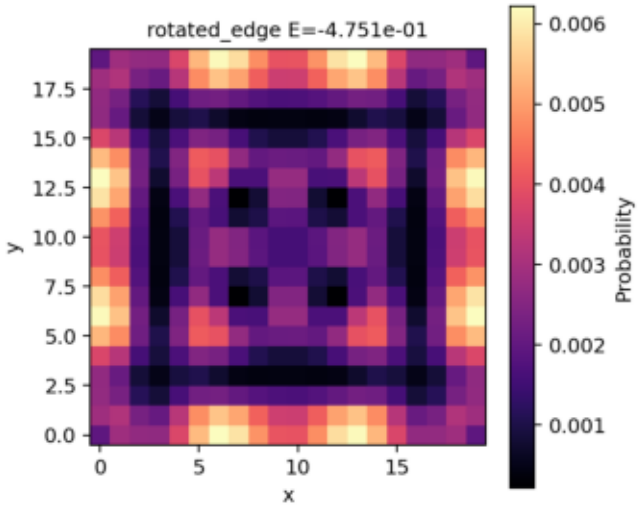
old



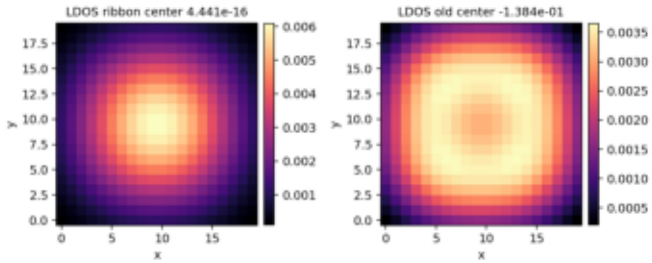
best_edge



rot_edge



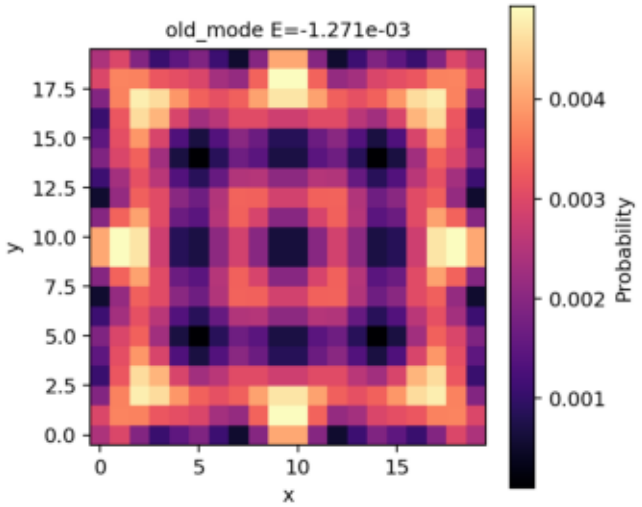
LDOS



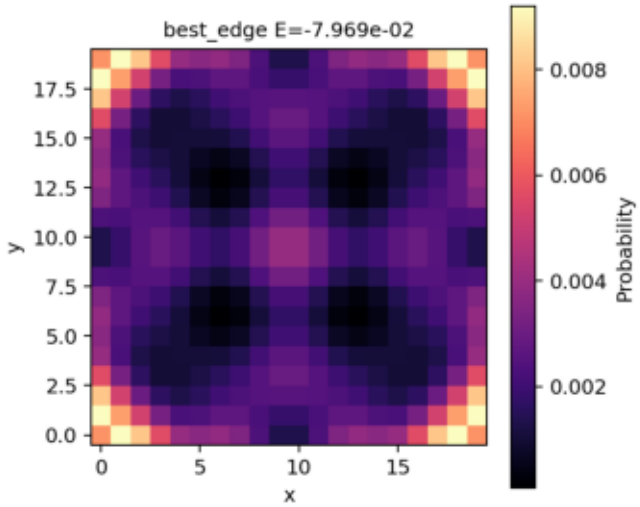
old_edge=0.139, best_edge=0.346, rot_edge=0.572; role=edge_dominated

□□A - v_0p70_t_1p00_lm_0p60
v=0.70, t=1.00, lm=0.60, gap=2.000e-01, Z2=1, Chern=0.000, Wilson=0.959

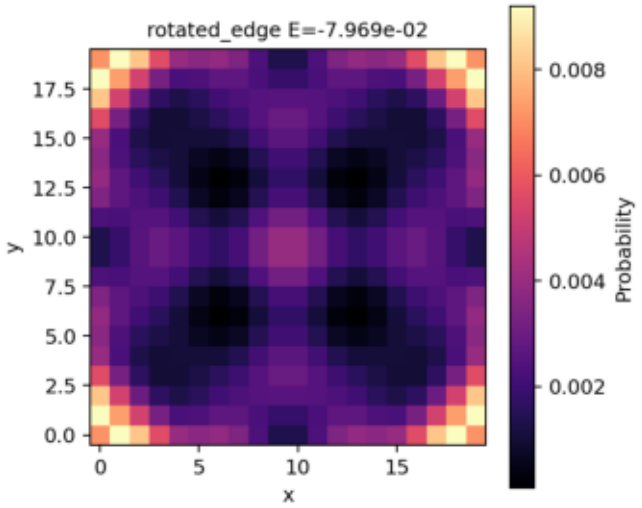
old



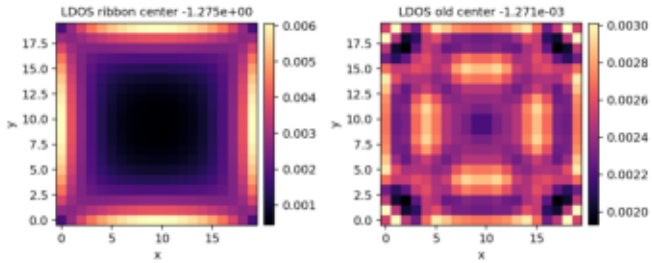
best_edge



rot_edge



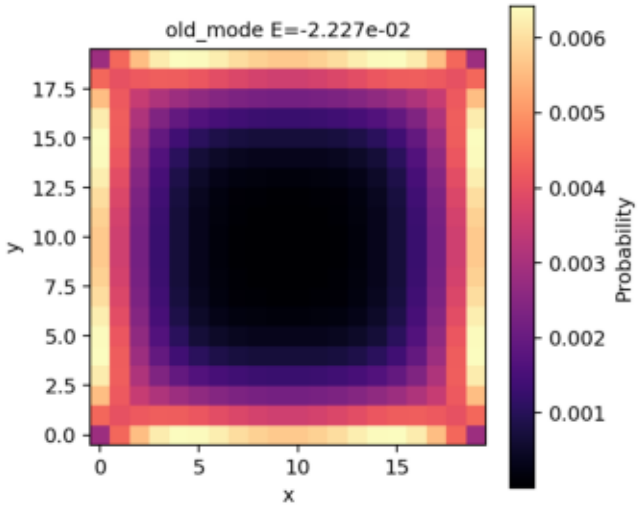
LDOS



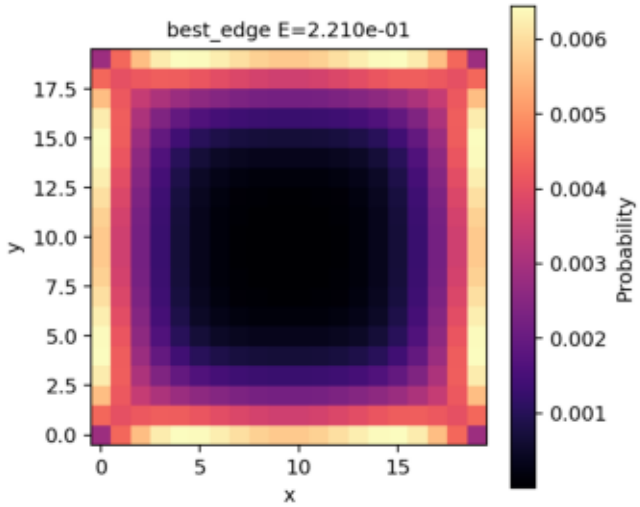
old_edge=0.365, best_edge=0.569, rot_edge=0.569; role=edge_dominated

□□A - v_0p60_t_0p70_lm_0p80
v=0.60, t=0.70, lm=0.80, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=-2.593

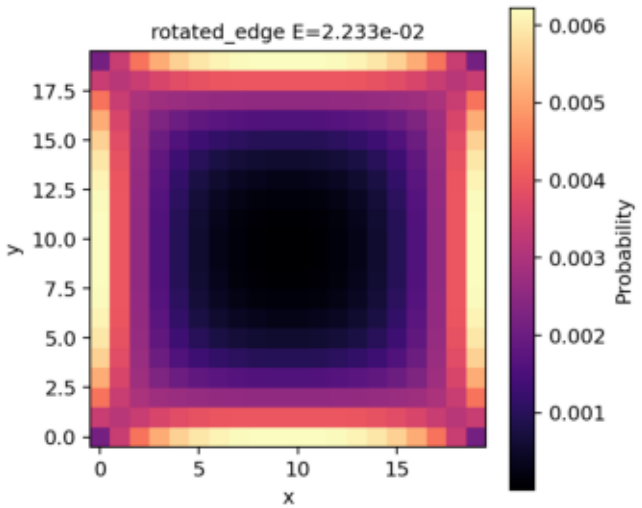
old



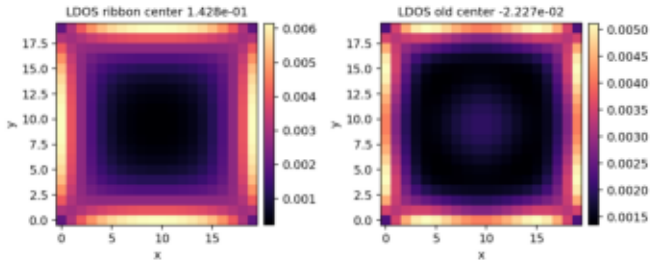
best_edge



rot_edge



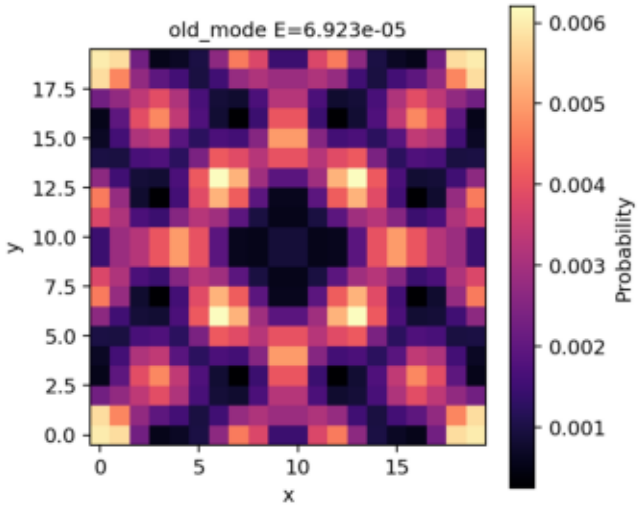
LDOS



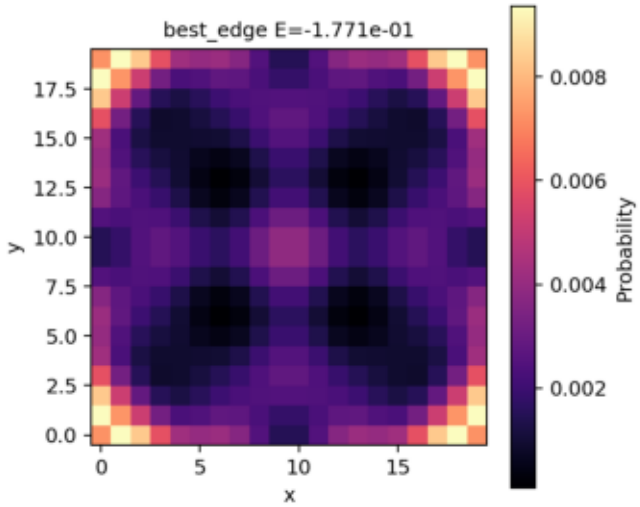
old_edge=0.703, best_edge=0.705, rot_edge=0.661; role=edge_dominated

□□A - v_0p60_t_1p00_lm_0p50
v=0.60, t=1.00, lm=0.50, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=0.511

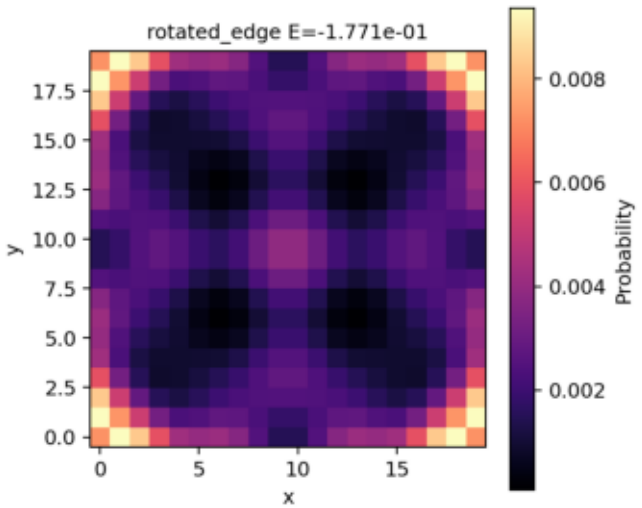
old



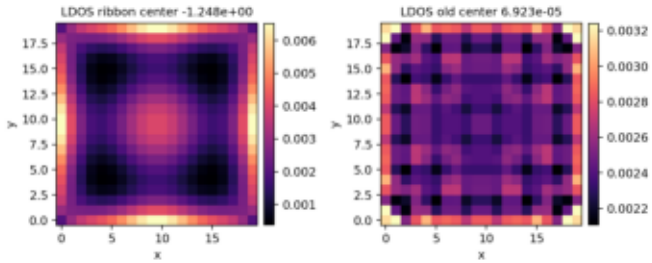
best_edge



rot_edge



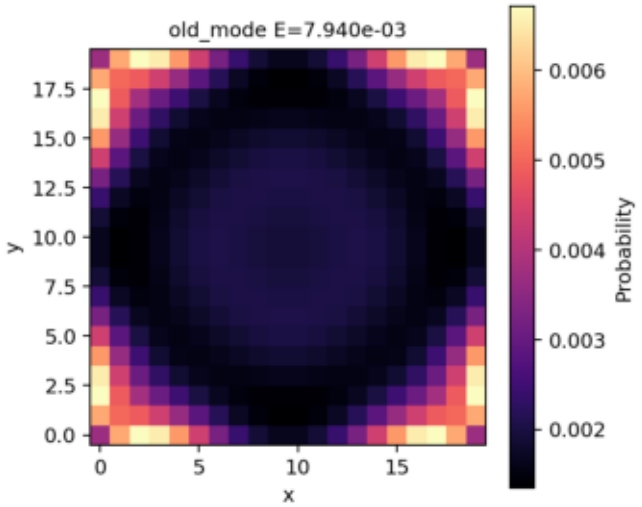
LDOS



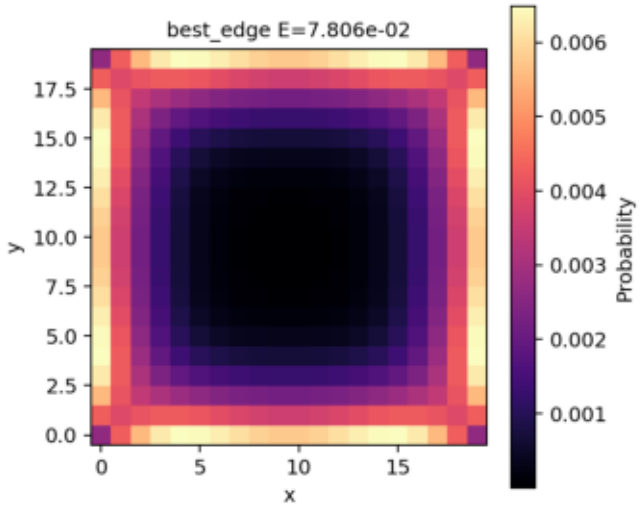
old_edge=0.364, best_edge=0.587, rot_edge=0.587; role=edge_dominated

□□A - v_0p30_t_0p50_lm_0p70
v=0.30, t=0.50, lm=0.70, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=-1.005

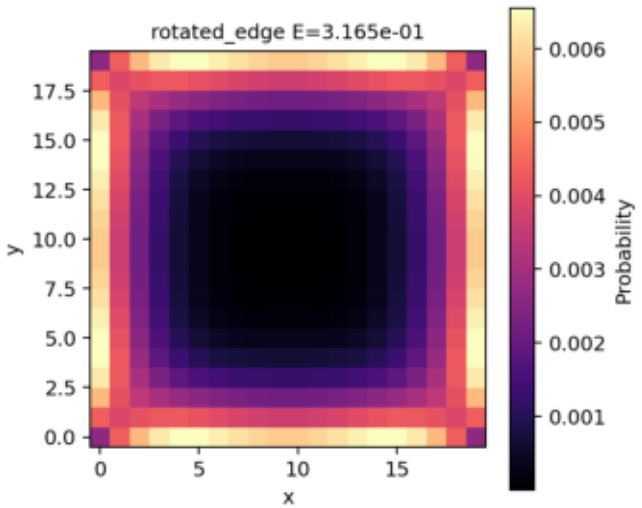
old



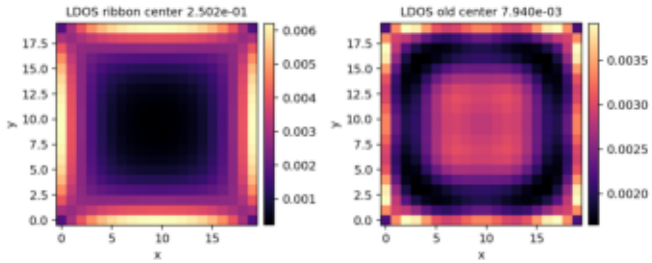
best_edge



rot_edge



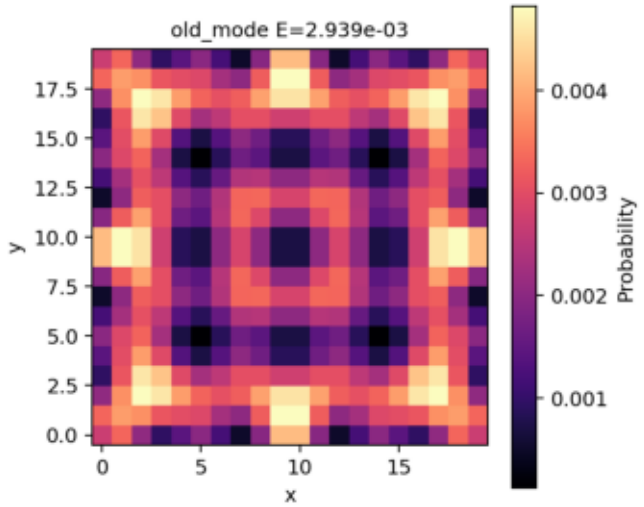
LDOS



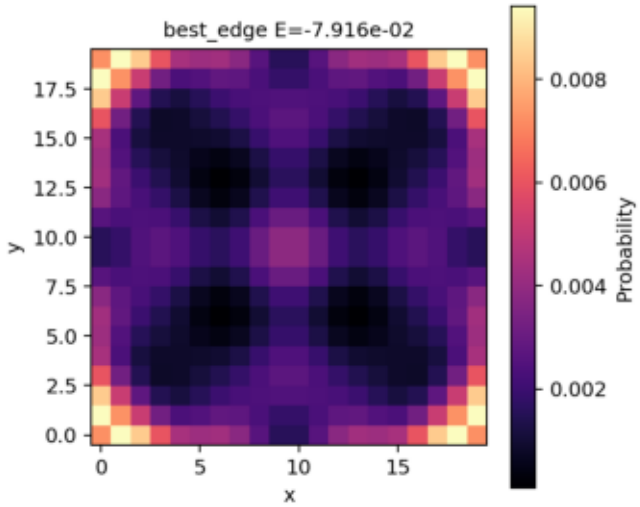
old_edge=0.519, best_edge=0.705, rot_edge=0.710; role=edge_dominated

□□A - v_0p50_t_0p90_lm_0p50
v=0.50, t=0.90, lm=0.50, gap=2.000e-01, Z2=1, Chern=0.000, Wilson=0.579

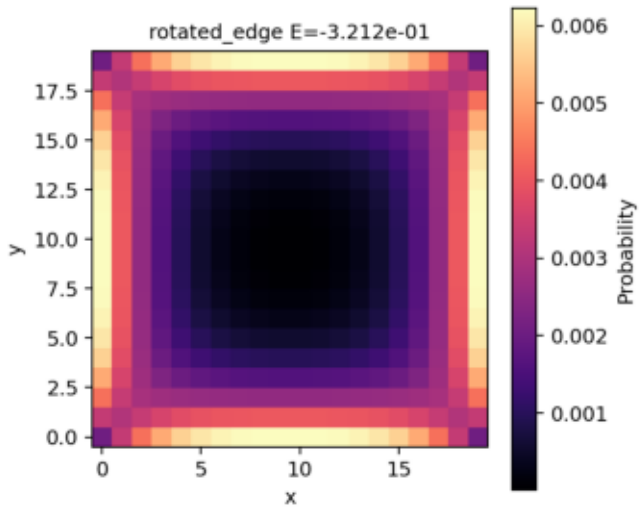
old



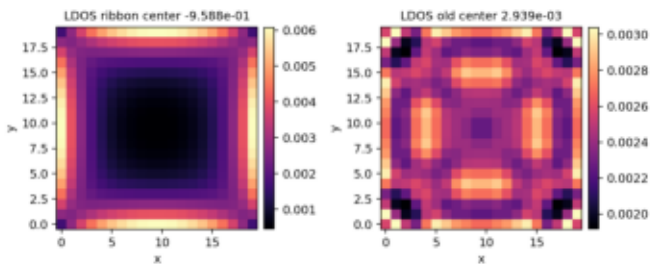
best_edge



rot_edge



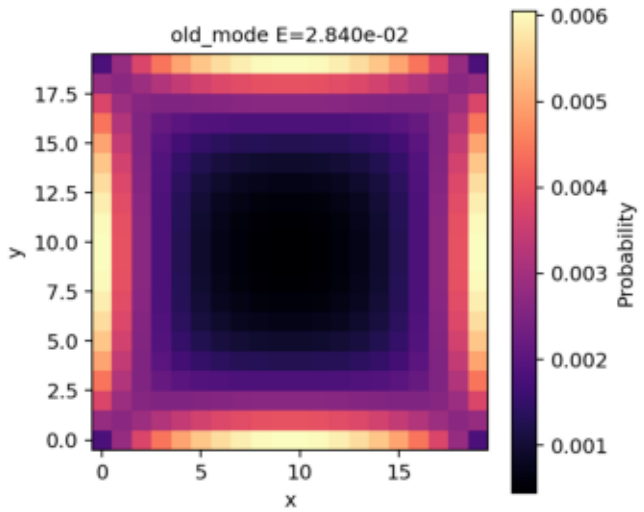
LDOS



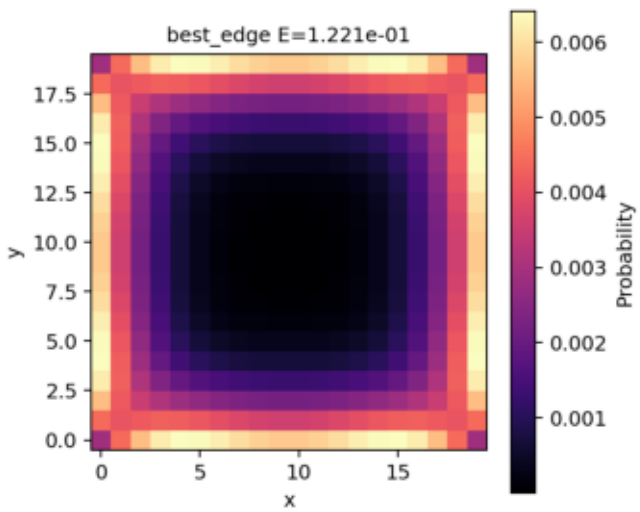
old_edge=0.367, best_edge=0.598, rot_edge=0.659; role=edge_dominated

□□A - v_0p70_t_0p80_lm_0p80
v=0.70, t=0.80, lm=0.80, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=0.785

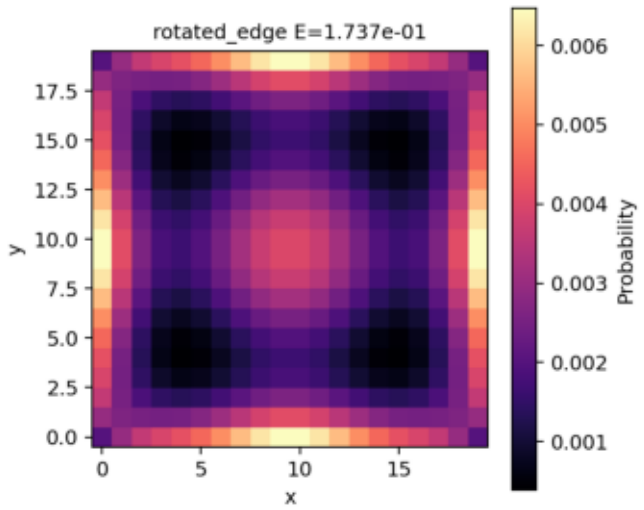
old



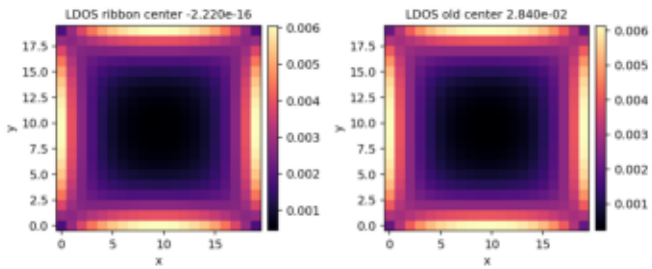
best_edge



rot_edge



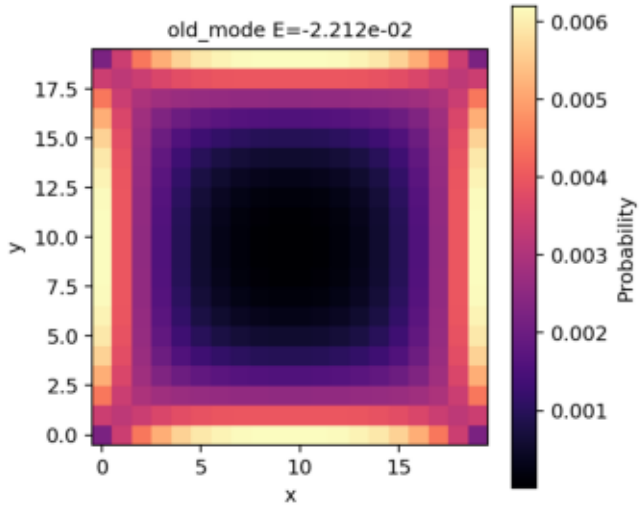
LDOS



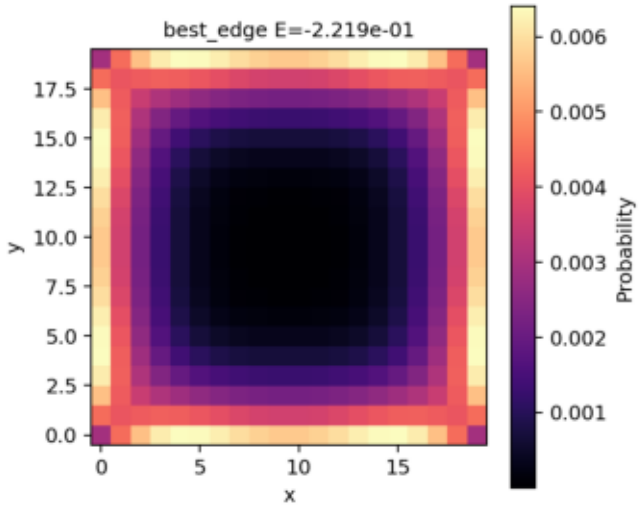
old_edge=0.607, best_edge=0.703, rot_edge=0.556; role=edge_dominated

□□A - v_0p80_t_0p90_lm_0p80
v=0.80, t=0.90, lm=0.80, gap=2.000e-01, Z2=1, Chern=-0.000, Wilson=0.553

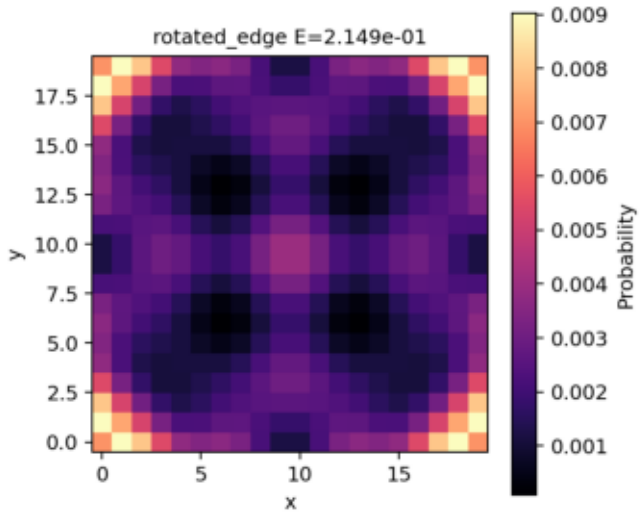
old



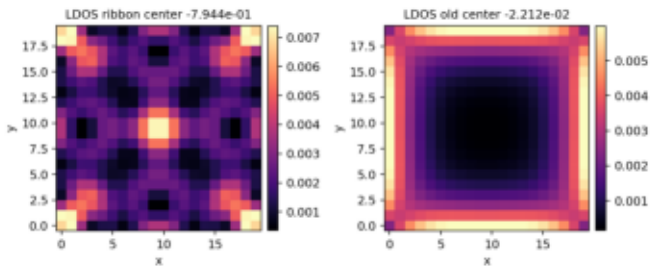
best_edge



rot_edge



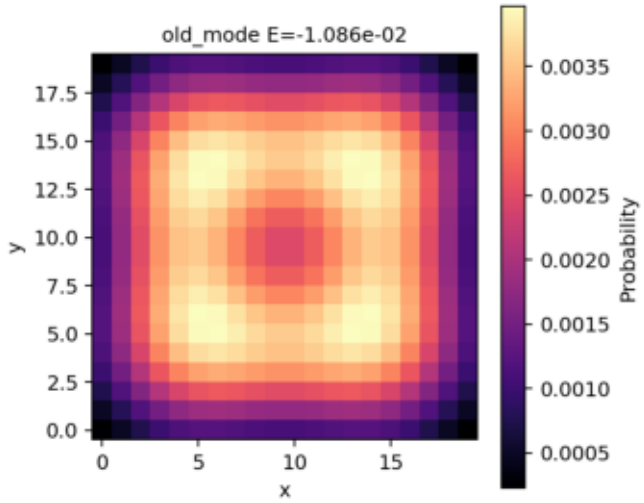
LDOS



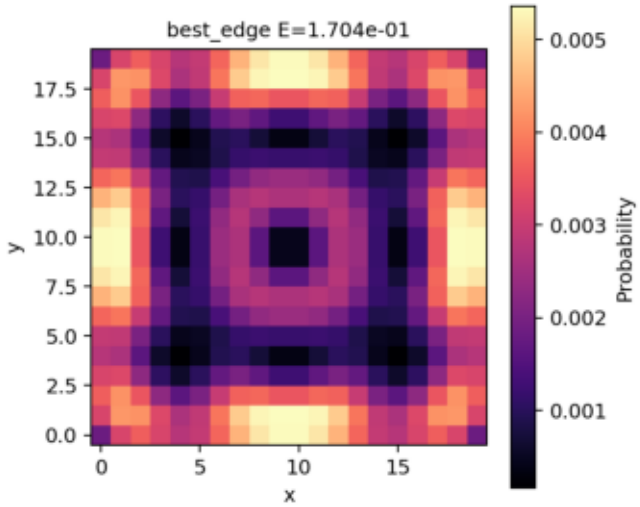
old_edge=0.661, best_edge=0.703, rot_edge=0.554; role=edge_dominated

□□A - v_0p50_t_0p90_lm_0p70
v=0.50, t=0.90, lm=0.70, gap=2.000e-01, Z2=1, Chern=0.000, Wilson=0.825

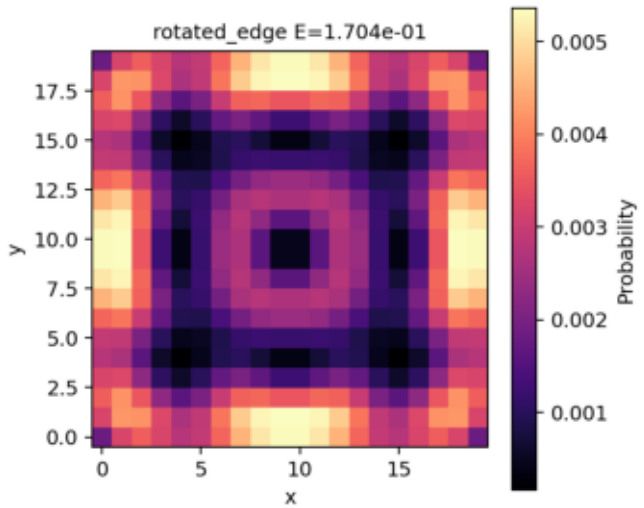
old



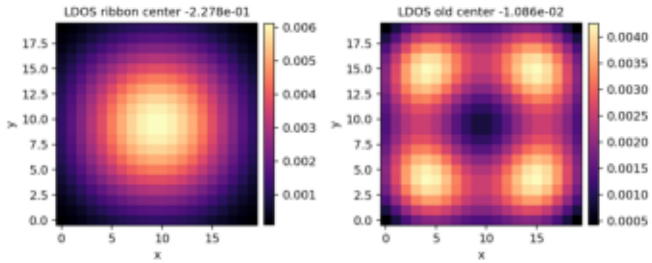
best_edge



rot_edge



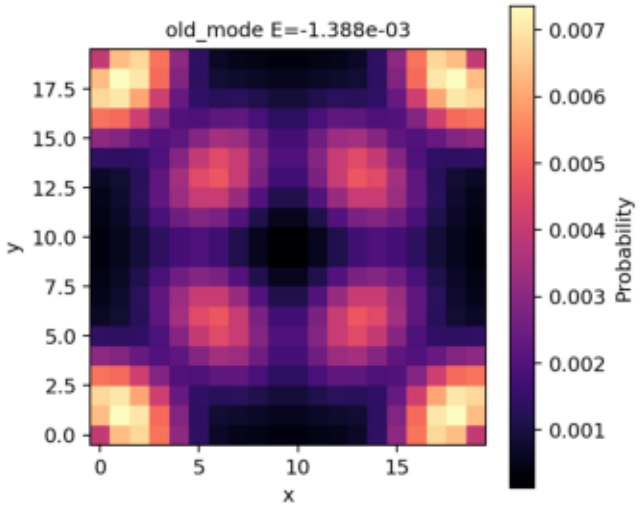
LDOS



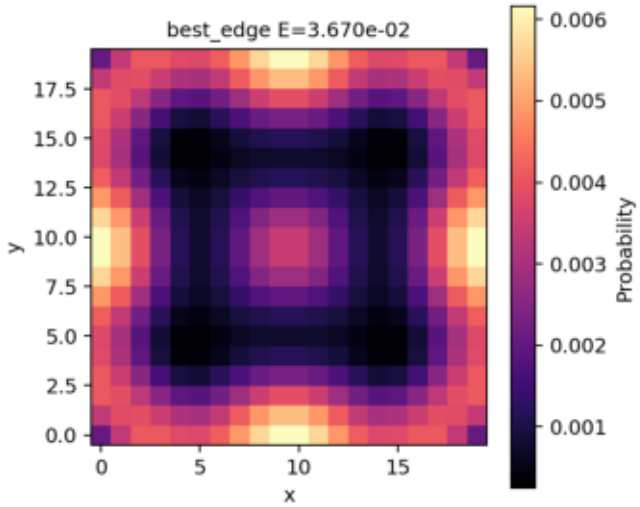
old_edge=0.187, best_edge=0.557, rot_edge=0.557; role=edge_dominated

□□A - v_0p50_t_0p60_lm_0p90
v=0.50, t=0.60, lm=0.90, gap=0.000e+00, Z2=1, Chern=0.000, Wilson=-1.093

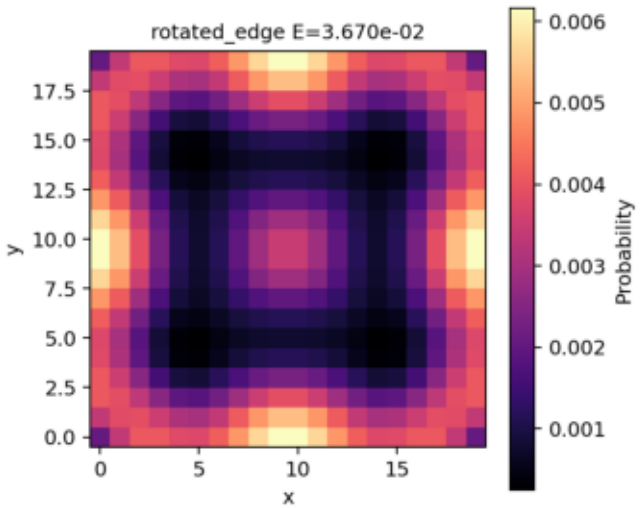
old



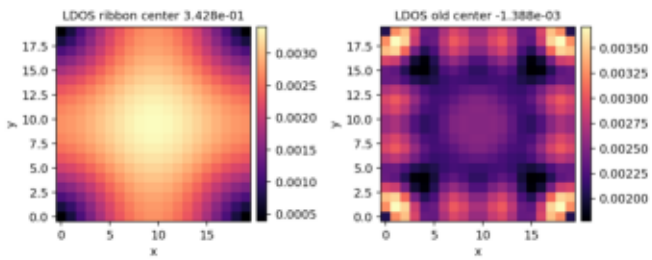
best_edge



rot_edge



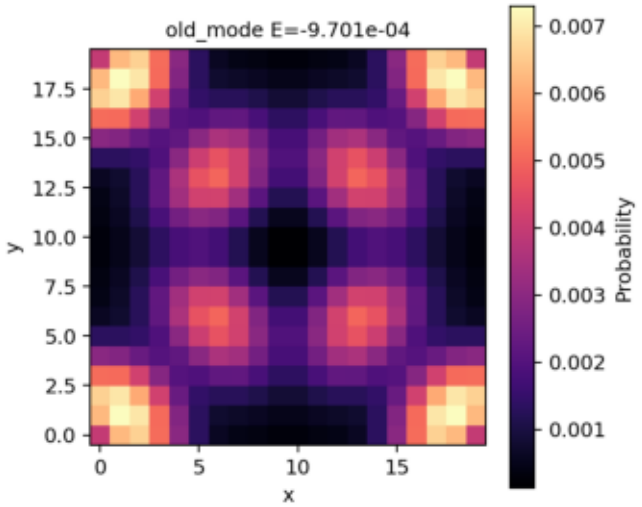
LDOS



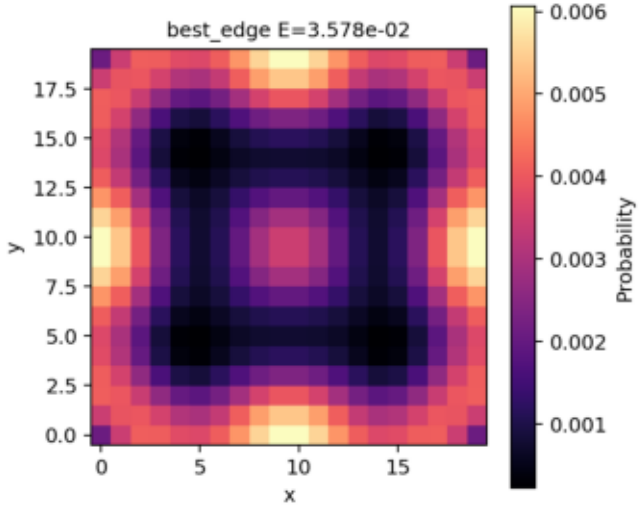
old_edge=0.396, best_edge=0.593, rot_edge=0.593; role=edge_dominated

□□A - v_0p70_t_0p70_lm_1p00
v=0.70, t=0.70, lm=1.00, gap=0.000e+00, Z2=0, Chern=0.000, Wilson=-0.254

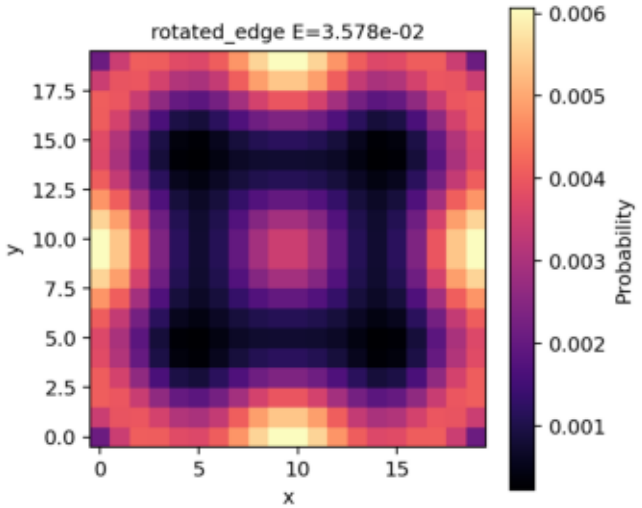
old



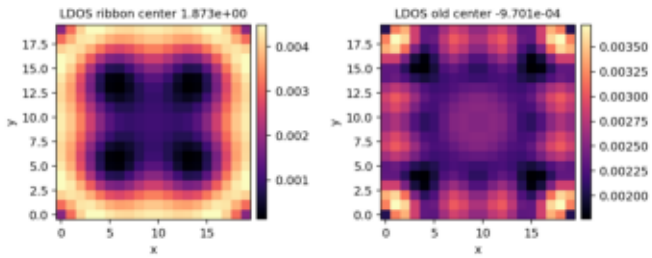
best_edge



rot_edge



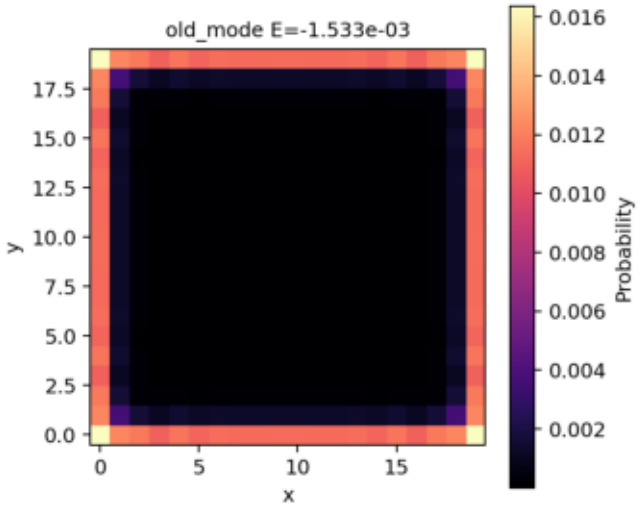
LDOS



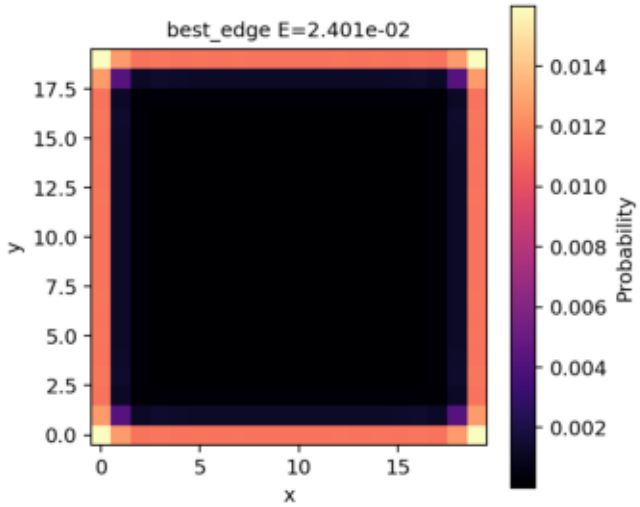
old_edge=0.386, best_edge=0.591, rot_edge=0.591; role=edge_dominated

□□A - v_0p50_t_0p20_lm_0p30
v=0.50, t=0.20, lm=0.30, gap=2.776e-17, Z2=1, Chern=-0.000, Wilson=0.552

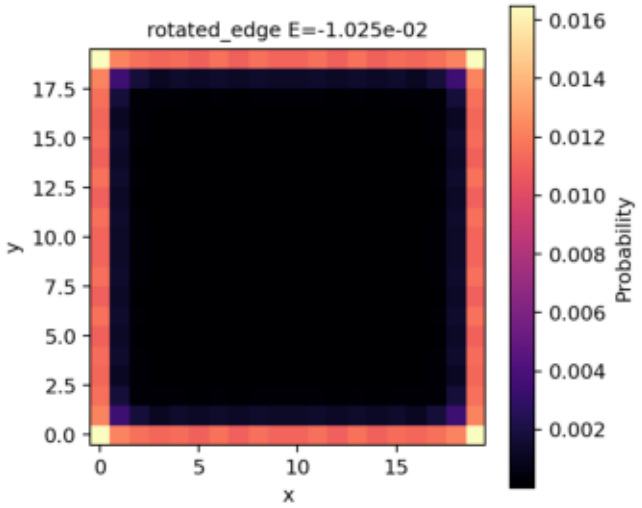
old



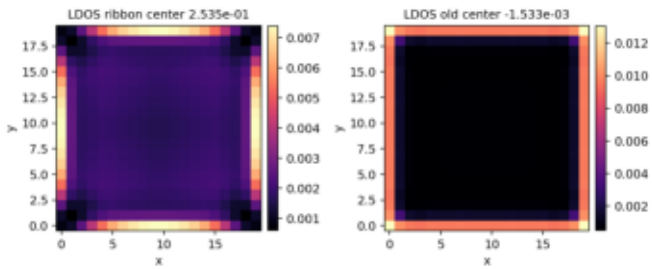
best_edge



rot_edge



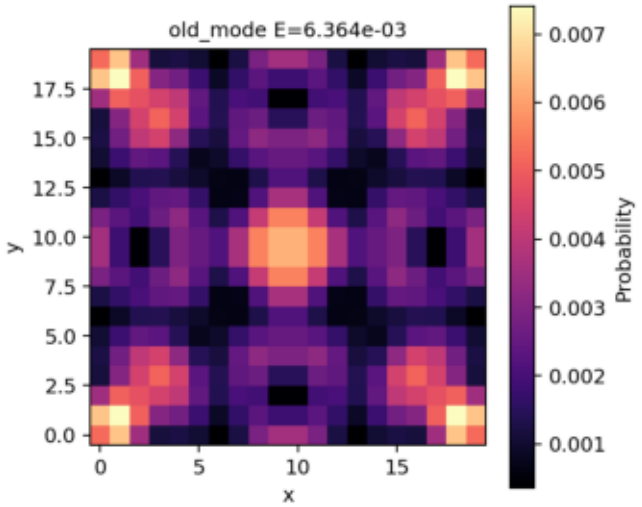
LDOS



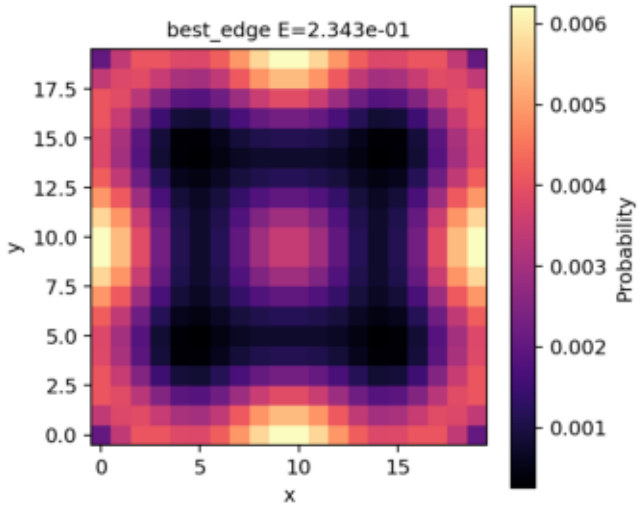
old_edge=0.989, best_edge=0.989, rot_edge=0.986; role=edge_dominated

□□A - v_0p50_t_0p50_lm_1p00
v=0.50, t=0.50, lm=1.00, gap=5.551e-17, Z2=1, Chern=0.000, Wilson=-1.058

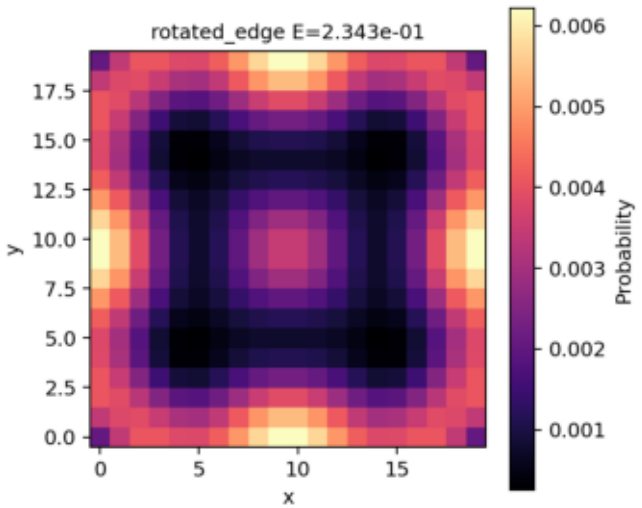
old



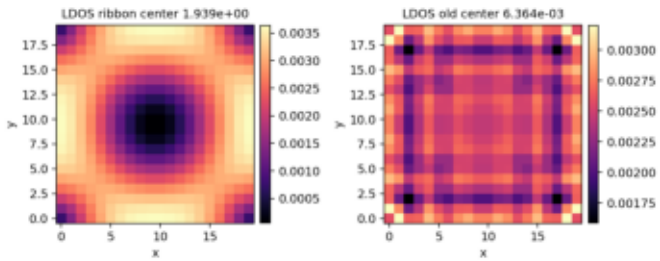
best_edge



rot_edge



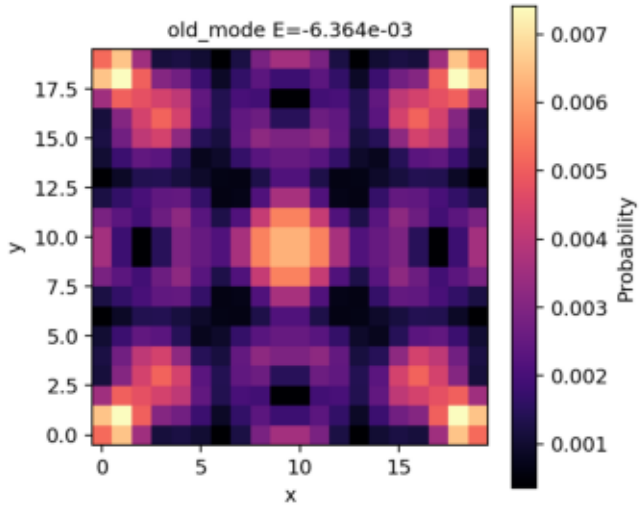
LDOS



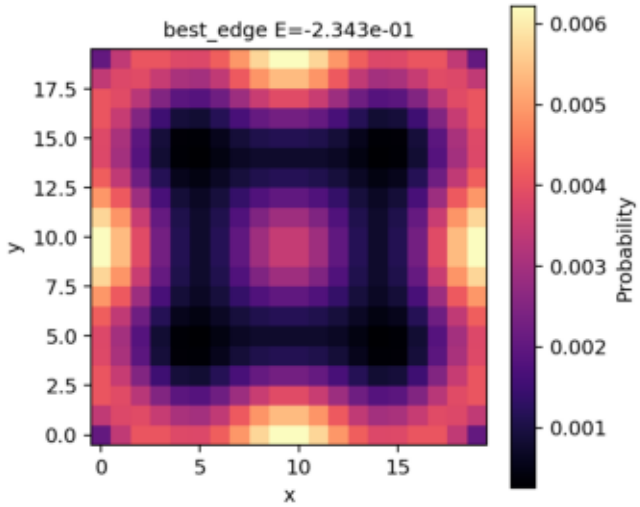
old_edge=0.376, best_edge=0.594, rot_edge=0.594; role=edge_dominated

□□A - v_0p50_t_1p00_lm_0p50
v=0.50, t=1.00, lm=0.50, gap=-5.551e-17, Z2=1, Chern=0.000, Wilson=-0.605

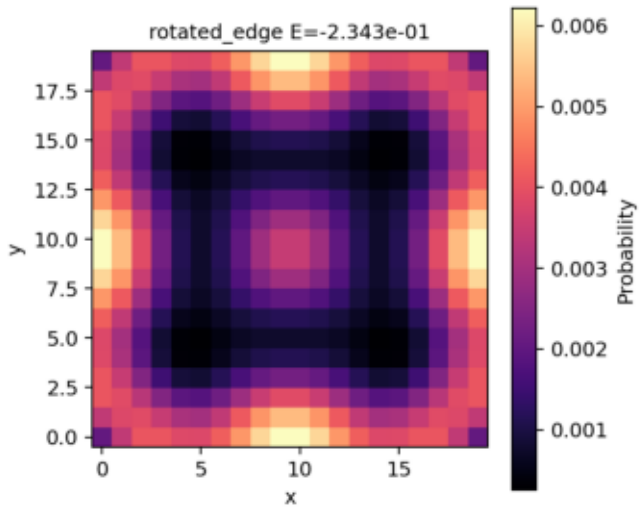
old



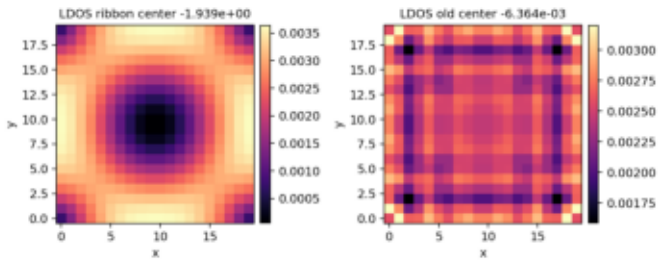
best_edge



rot_edge



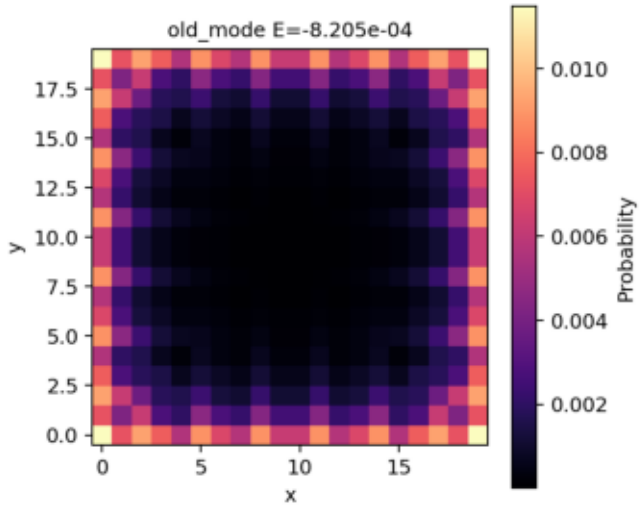
LDOS



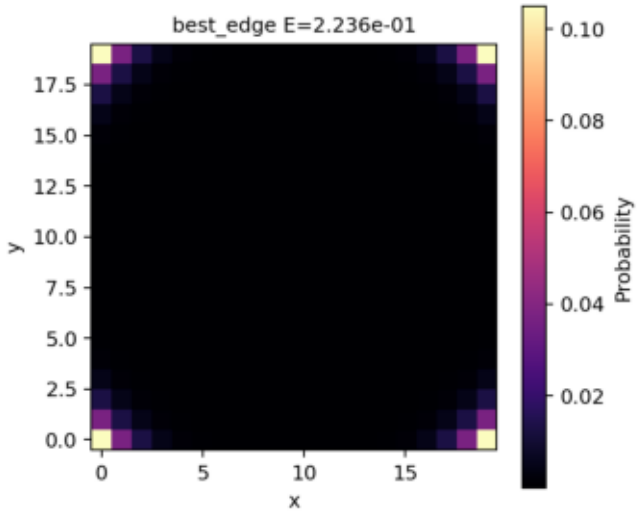
old_edge=0.376, best_edge=0.594, rot_edge=0.594; role=edge_dominated

□□A - v_0p70_t_0p10_lm_0p20
v=0.70, t=0.10, lm=0.20, gap=5.551e-17, Z2=0, Chern=-0.000, Wilson=-0.049

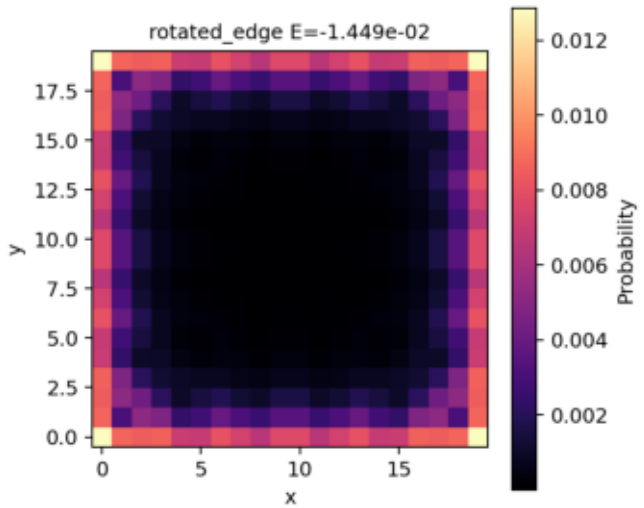
old



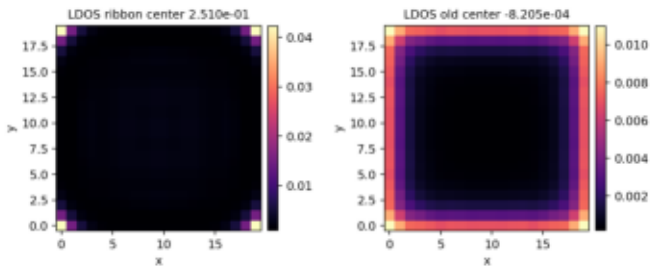
best_edge



rot_edge



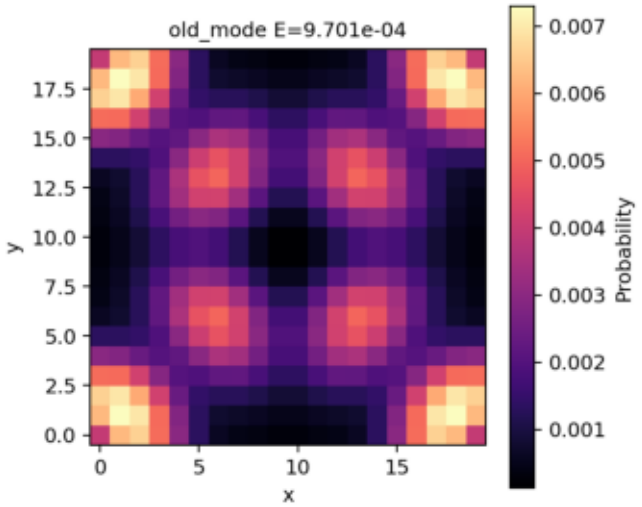
LDOS



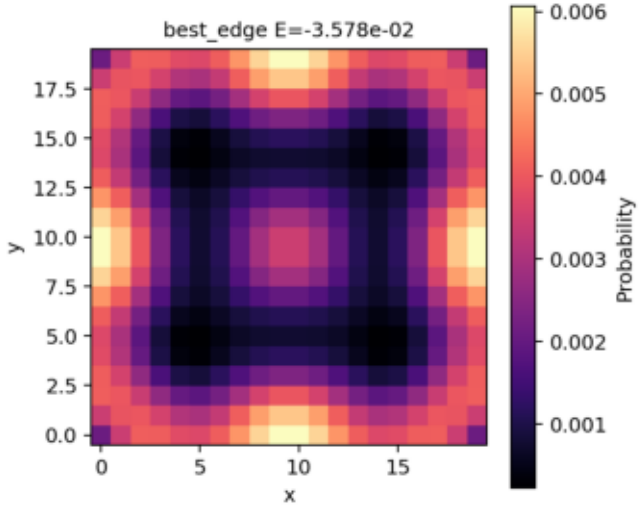
old_edge=0.810, best_edge=0.985, rot_edge=0.841; role=corner_dominated

□□A - v_0p70_t_1p00_lm_0p70
v=0.70, t=1.00, lm=0.70, gap=-5.551e-17, Z2=1, Chern=0.000, Wilson=-0.537

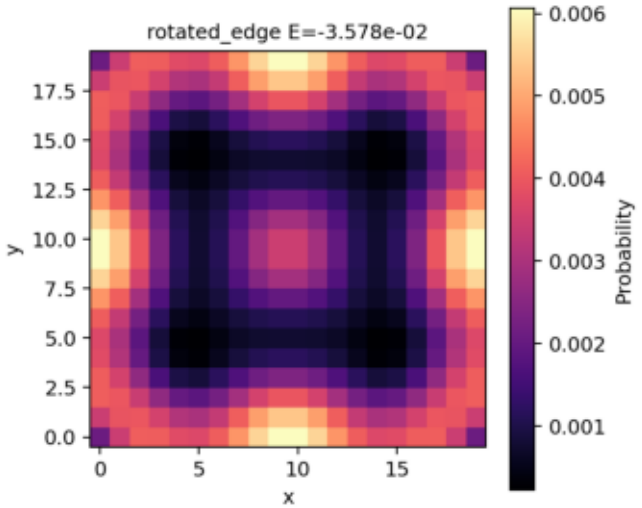
old



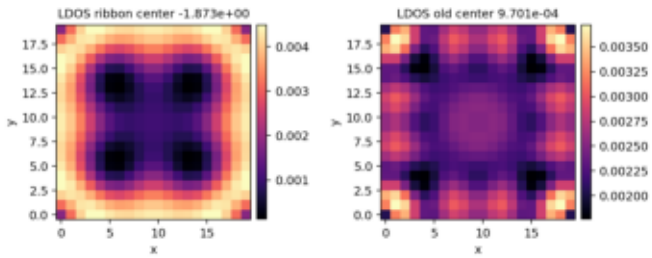
best_edge



rot_edge



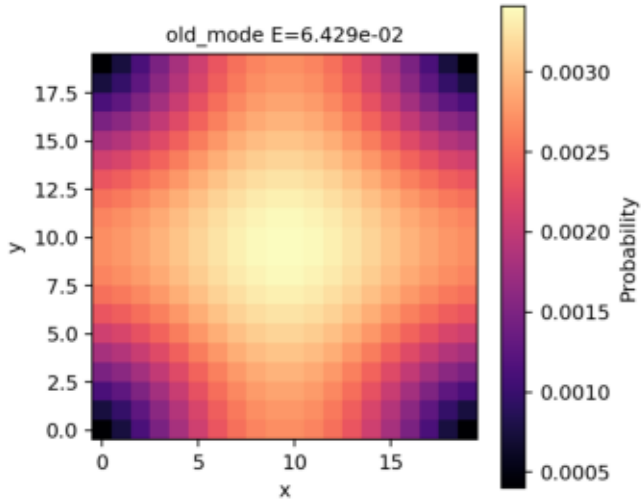
LDOS



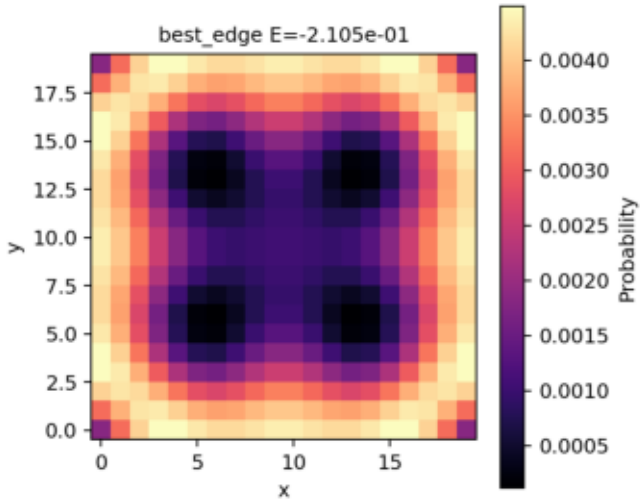
old_edge=0.386, best_edge=0.591, rot_edge=0.591; role=edge_dominated

□□A - v_0p60_t_0p80_lm_0p80
v=0.60, t=0.80, lm=0.80, gap=-6.221e-17, Z2=0, Chern=-1.000, Wilson=0.458

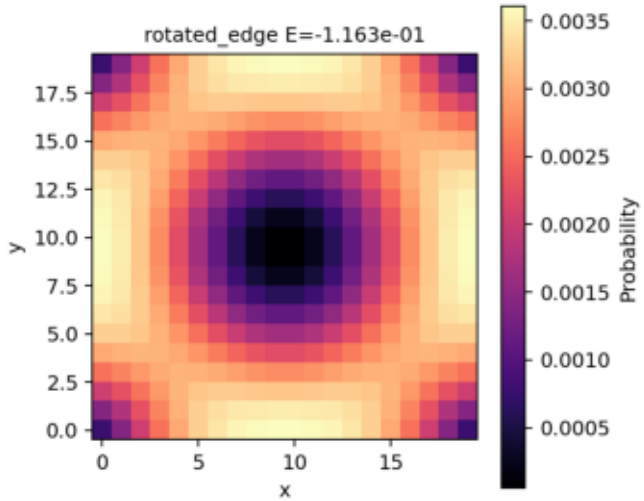
old



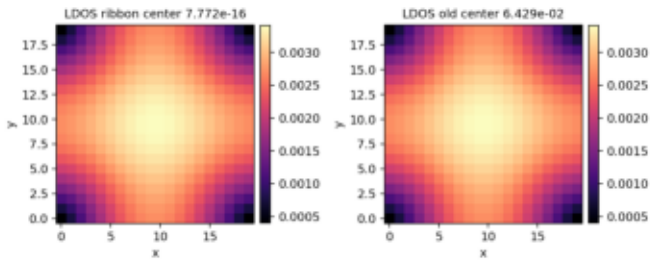
best_edge



rot_edge



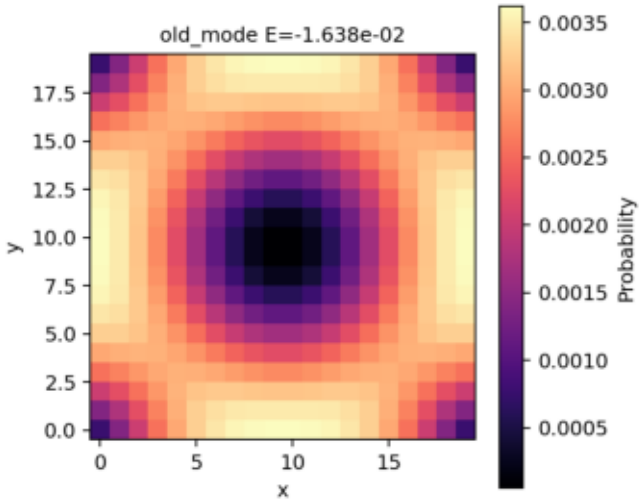
LDOS



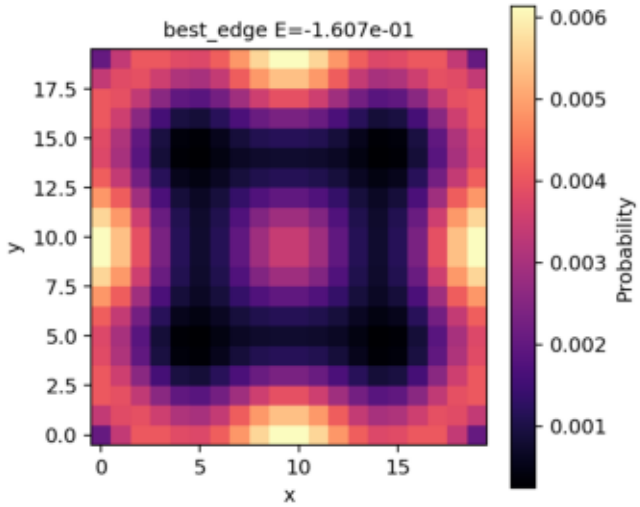
old_edge=0.283, best_edge=0.574, rot_edge=0.421; role=bulk_mixed

□□A - v_0p50_t_0p70_lm_0p80
v=0.50, t=0.70, lm=0.80, gap=6.939e-17, Z2=0, Chern=-0.000, Wilson=0.482

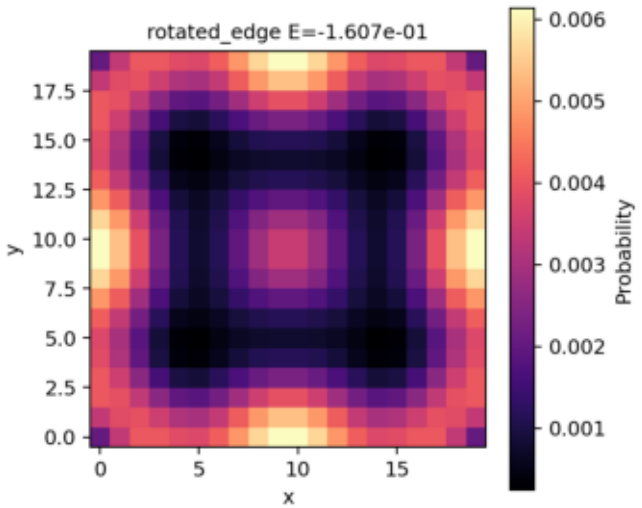
old



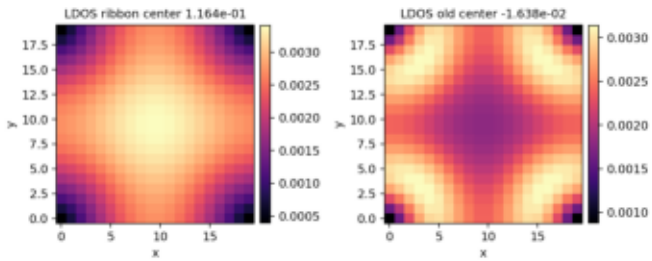
best_edge



rot_edge



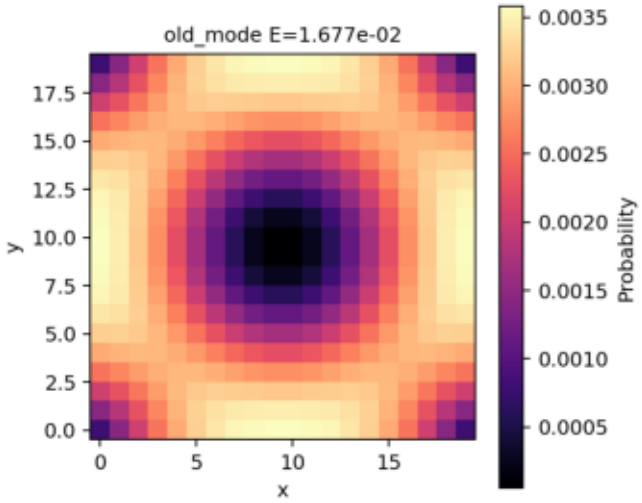
LDOS



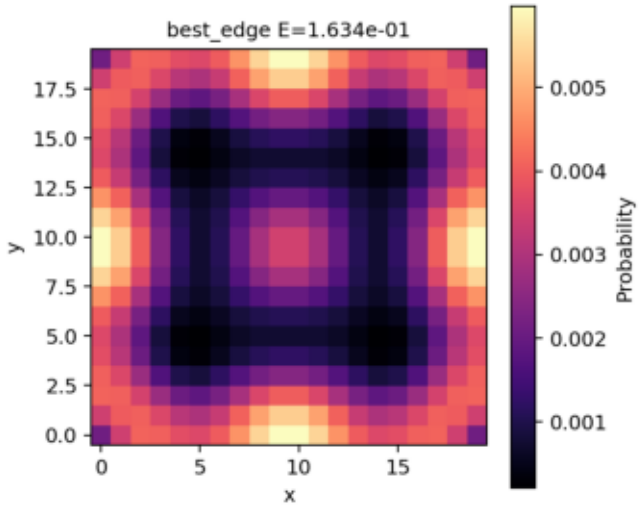
old_edge=0.422, best_edge=0.592, rot_edge=0.592; role=edge_dominated

□□A - v_0p90_t_1p00_lm_0p90
v=0.90, t=1.00, lm=0.90, gap=-6.939e-17, Z2=0, Chern=-0.000, Wilson=-0.250

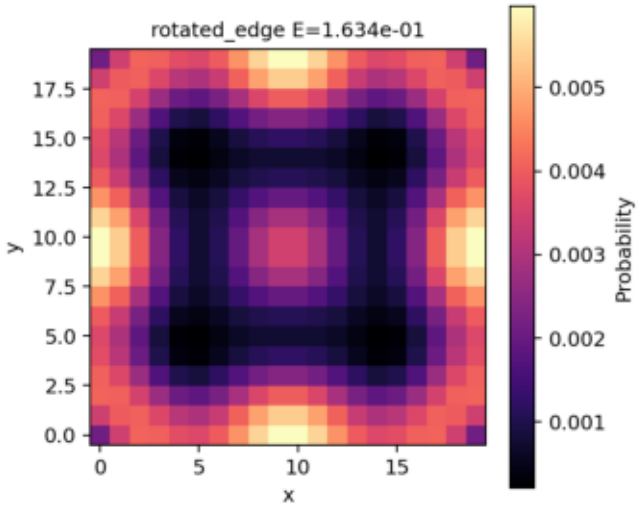
old



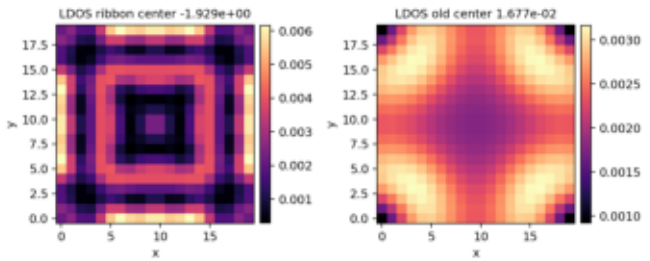
best_edge



rot_edge



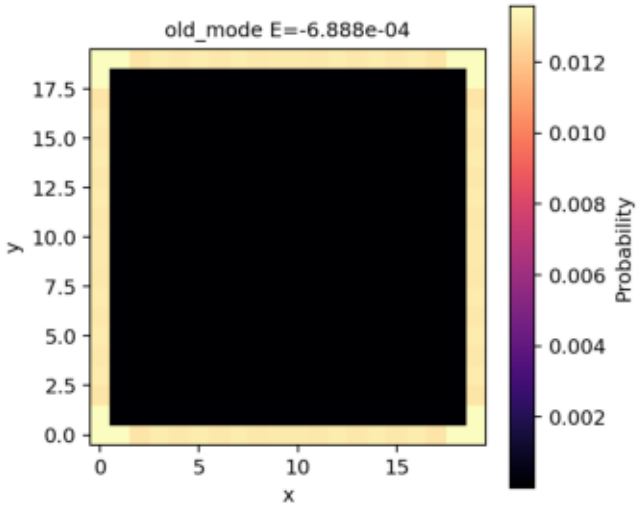
LDOS



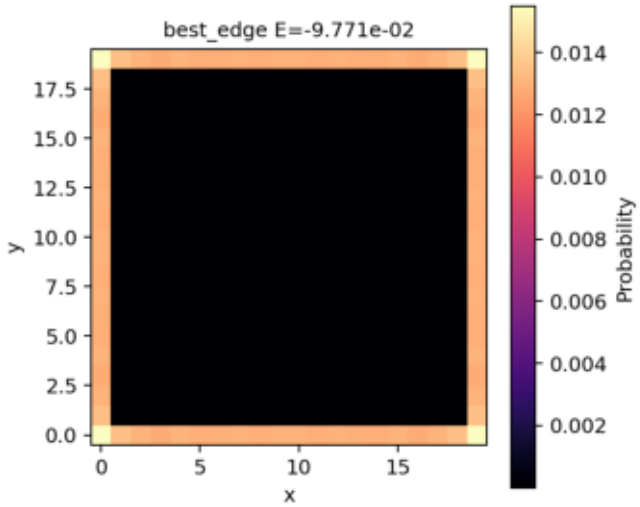
old_edge=0.420, best_edge=0.588, rot_edge=0.588; role=edge_dominated

□□A - v_0p30_t_0p30_lm_0p40
v=0.30, t=0.30, lm=0.40, gap=8.327e-17, Z2=1, Chern=-0.000, Wilson=0.985

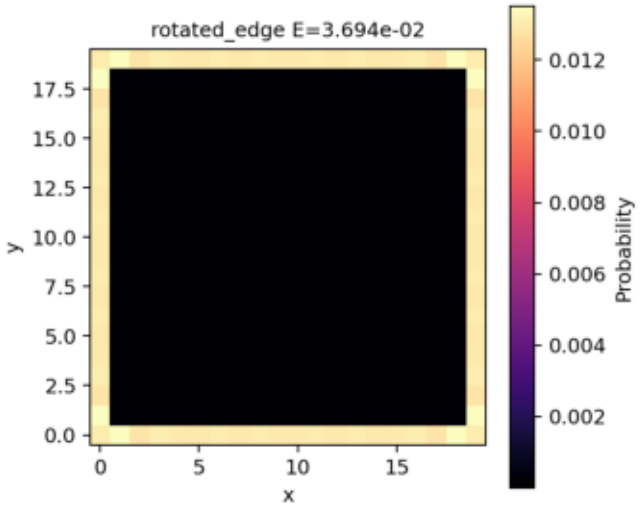
old



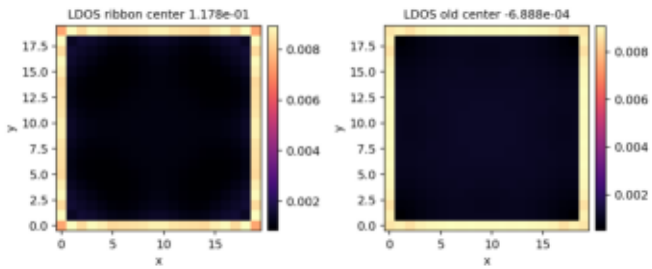
best_edge



rot_edge



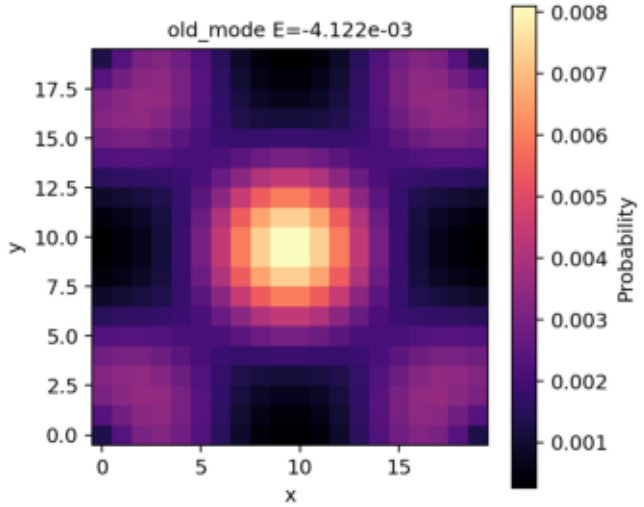
LDOS



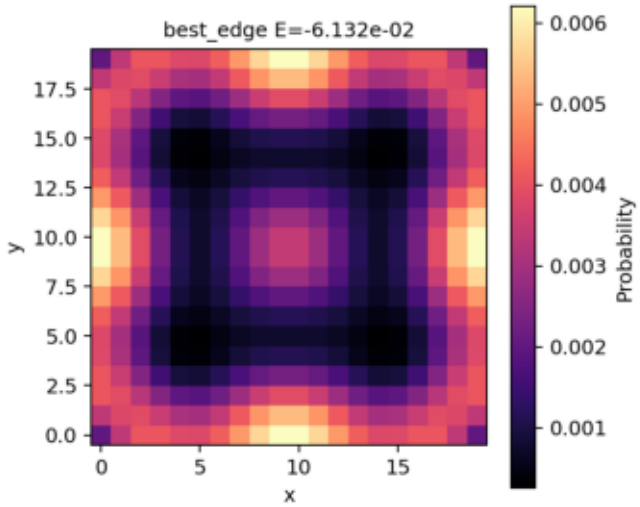
old_edge=0.999, best_edge=1.000, rot_edge=0.996; role=corner_dominated

□□A - v_0p40_t_0p60_lm_0p80
v=0.40, t=0.60, lm=0.80, gap=8.327e-17, Z2=0, Chern=-0.000, Wilson=0.008

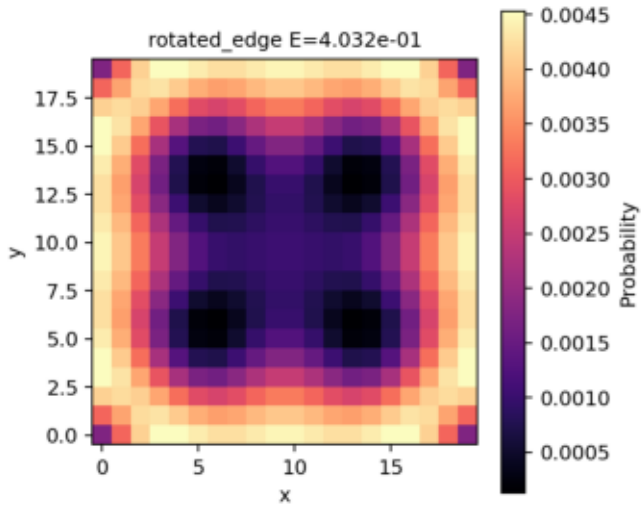
old



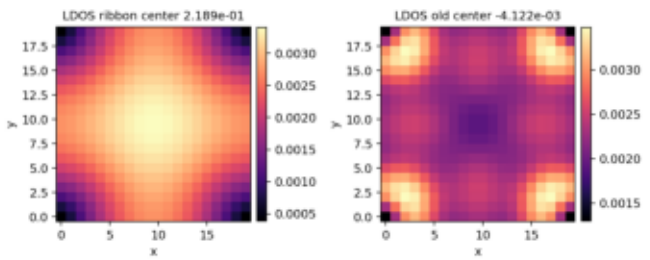
best_edge



rot_edge



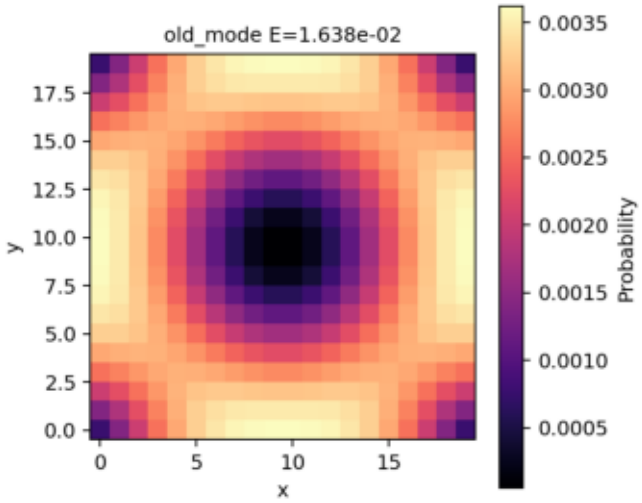
LDOS



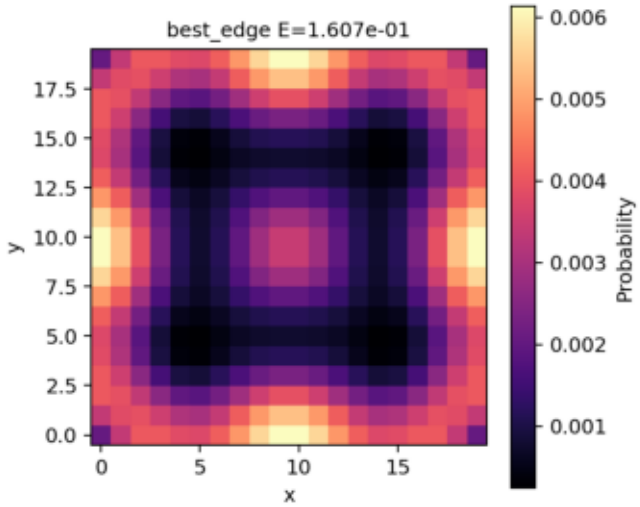
old_edge=0.277, best_edge=0.594, rot_edge=0.578; role=edge_dominated

□□A - v_0p50_t_0p80_lm_0p70
v=0.50, t=0.80, lm=0.70, gap=-9.714e-17, Z2=0, Chern=0.000, Wilson=-0.232

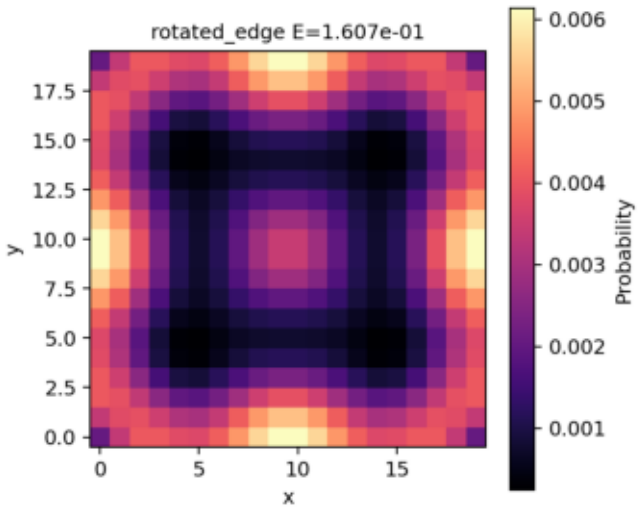
old



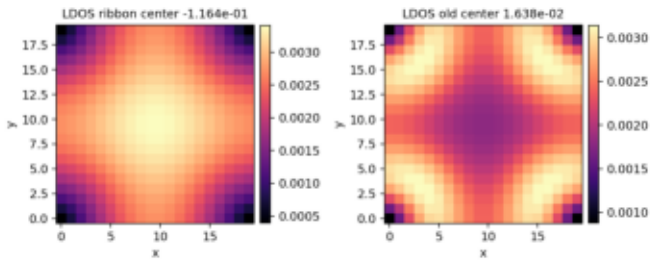
best_edge



rot_edge



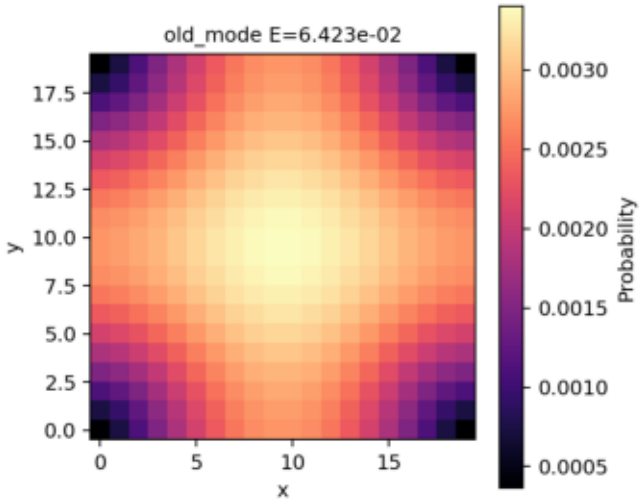
LDOS



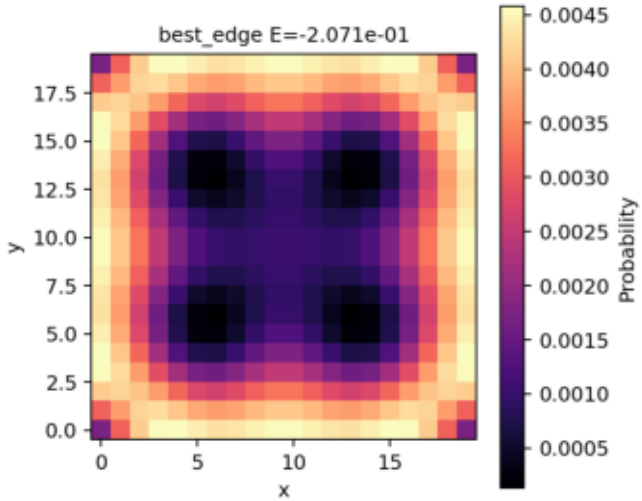
old_edge=0.422, best_edge=0.592, rot_edge=0.592; role=edge_dominated

□□A - v_0p20_t_0p60_lm_0p60
v=0.20, t=0.60, lm=0.60, gap=-1.050e-16, Z2=1, Chern=0.000, Wilson=0.800

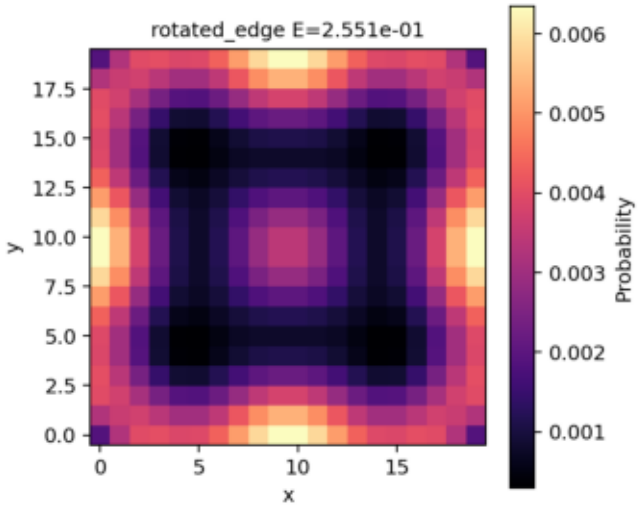
old



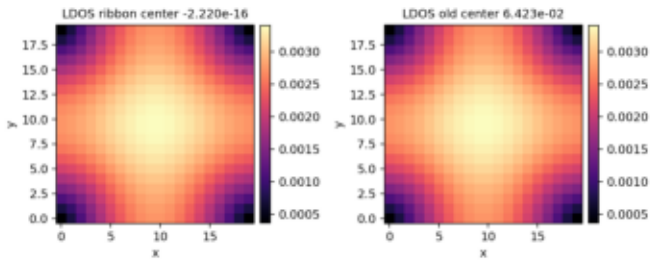
best_edge



rot_edge



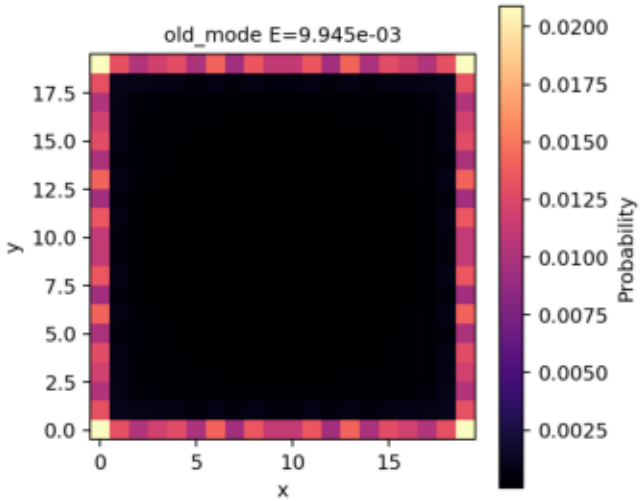
LDOS



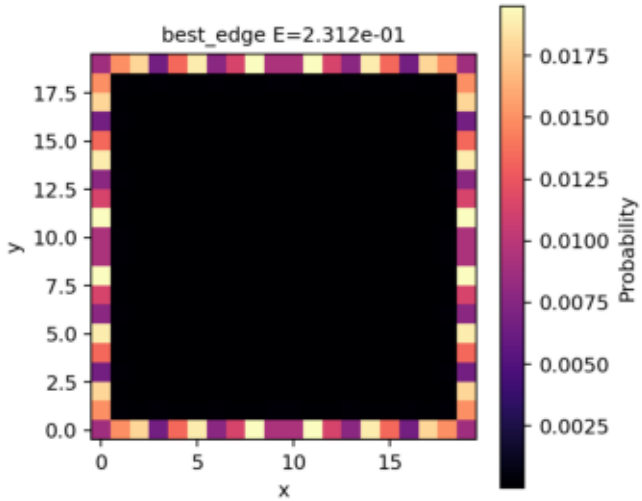
old_edge=0.284, best_edge=0.583, rot_edge=0.596; role=edge_dominated

□□A - v_0p30_t_0p10_lm_0p50
v=0.30, t=0.10, lm=0.50, gap=-1.110e-16, Z2=0, Chern=0.000, Wilson=-0.237

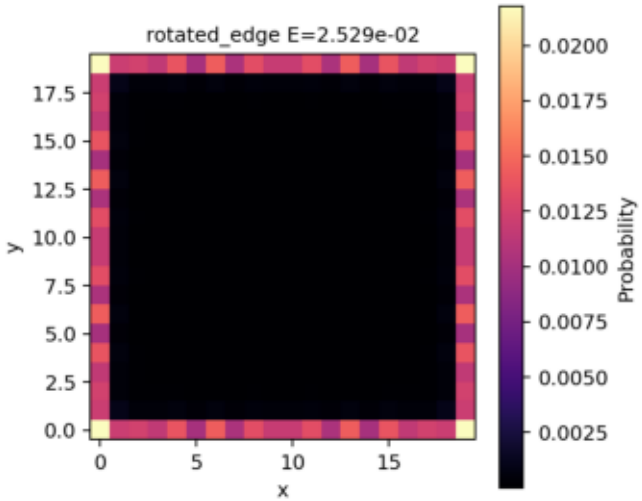
old



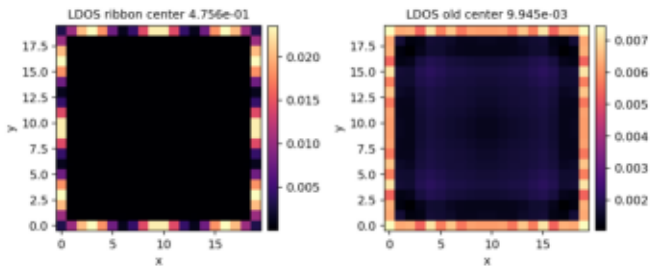
best_edge



rot_edge



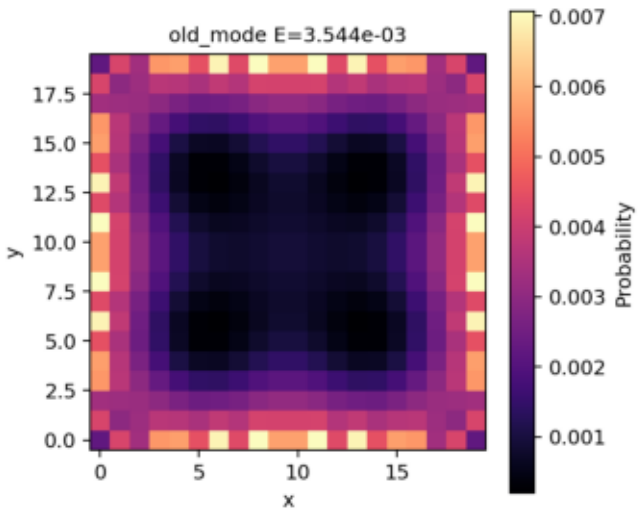
LDOS



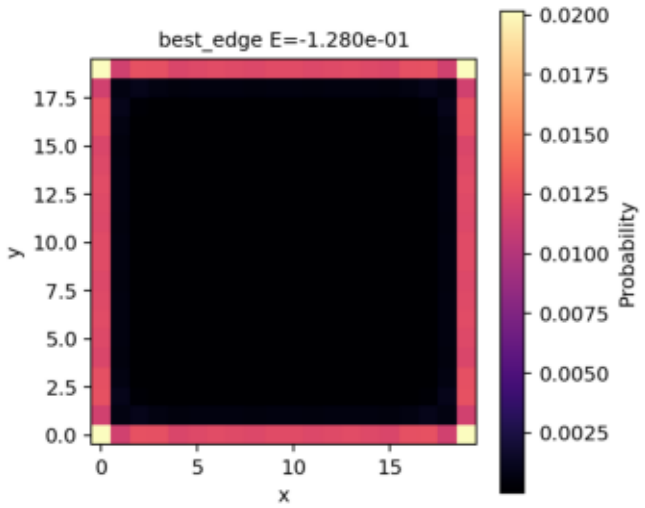
old_edge=0.979, best_edge=0.999, rot_edge=0.992; role=edge_dominated

□□A - v_0p40_t_0p20_lm_0p40
v=0.40, t=0.20, lm=0.40, gap=1.110e-16, Z2=0, Chern=0.000, Wilson=-0.256

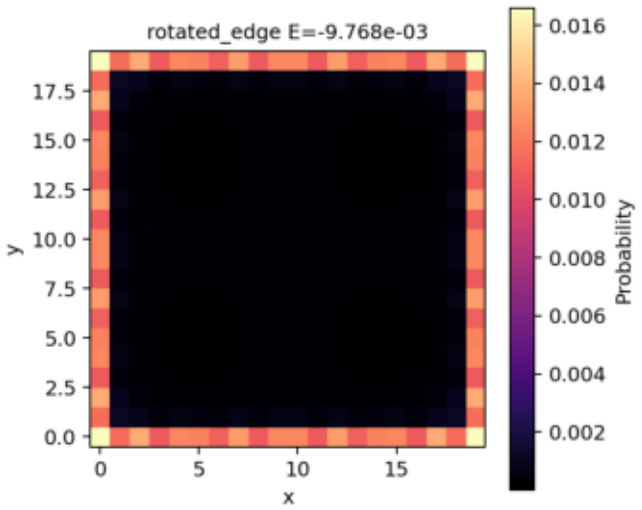
old



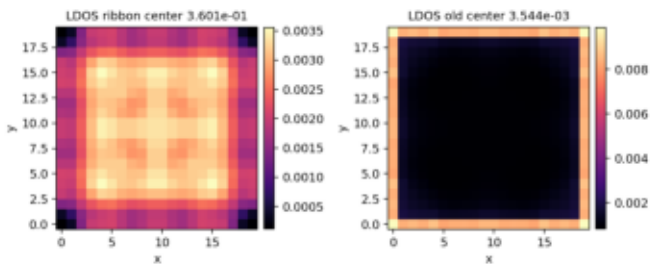
best_edge



rot_edge



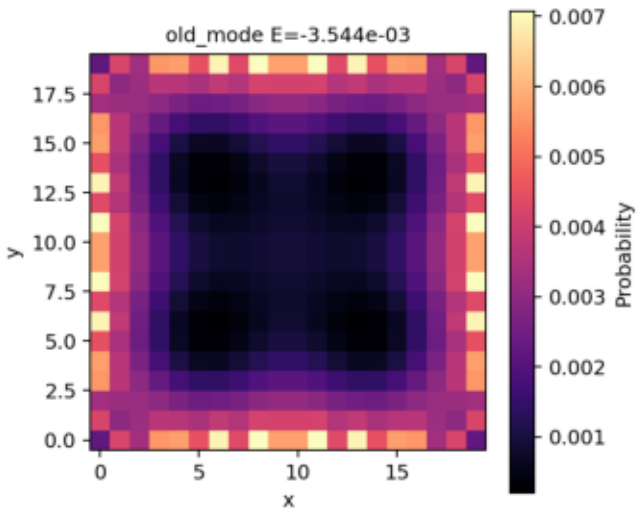
LDOS



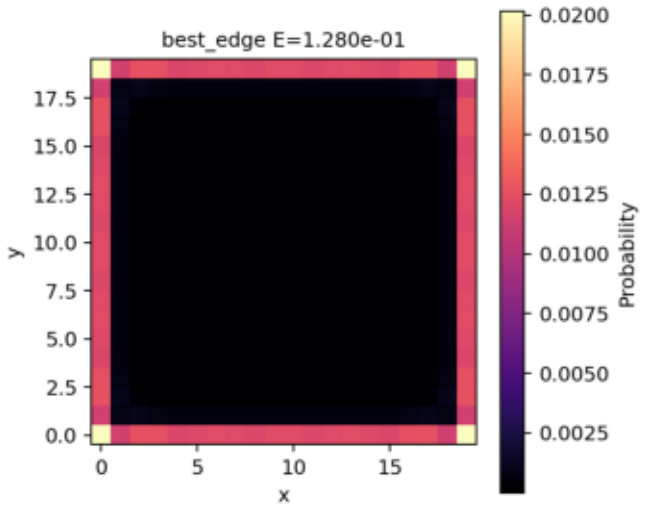
old_edge=0.637, best_edge=0.997, rot_edge=0.973; role=edge_dominated

□□A - v_0p40_t_0p40_lm_0p20
v=0.40, t=0.40, lm=0.20, gap=1.110e-16, Z2=0, Chern=0.000, Wilson=0.354

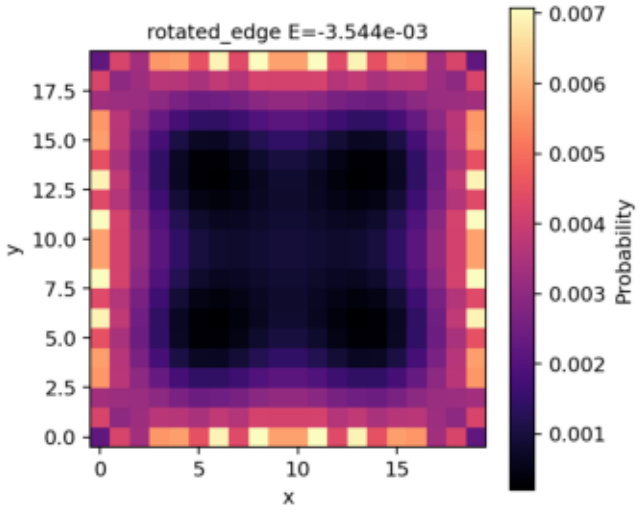
old



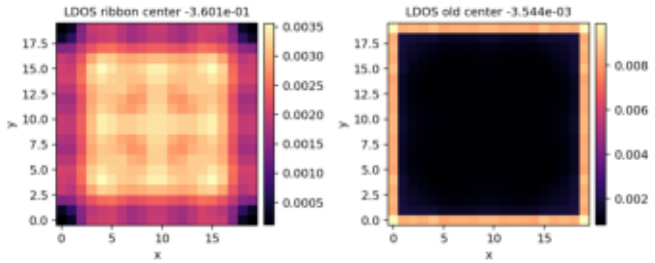
best_edge



rot_edge



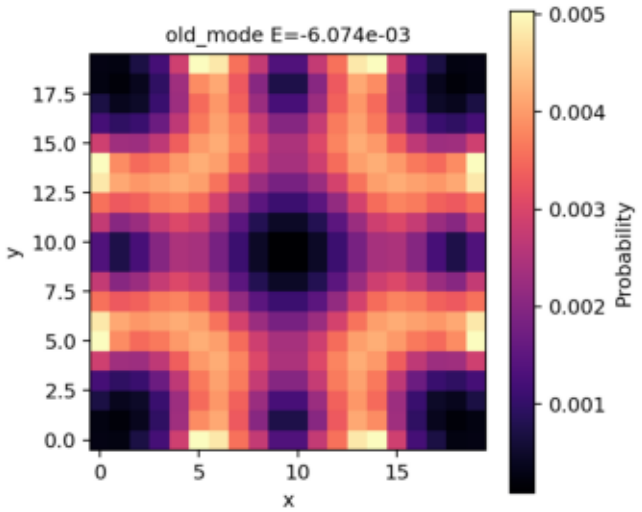
LDOS



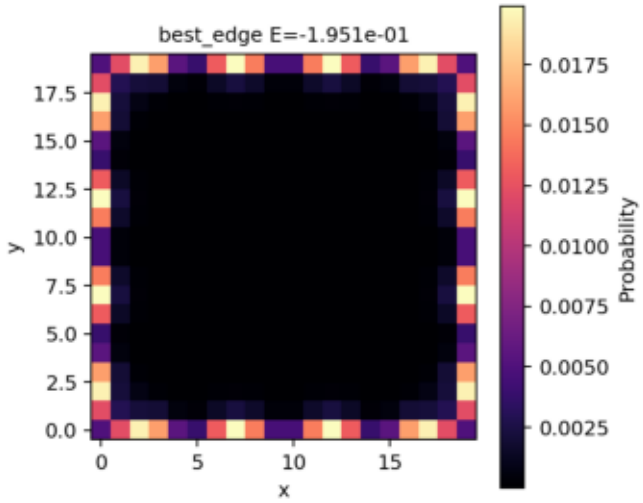
old_edge=0.637, best_edge=0.997, rot_edge=0.637; role=corner_dominated

□□A - v_0p50_t_0p40_lm_0p10
v=0.50, t=0.40, lm=0.10, gap=1.110e-16, Z2=1, Chern=0.000, Wilson=-0.756

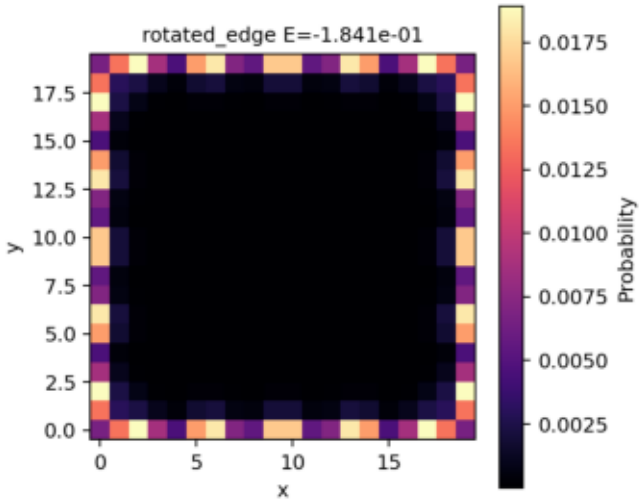
old



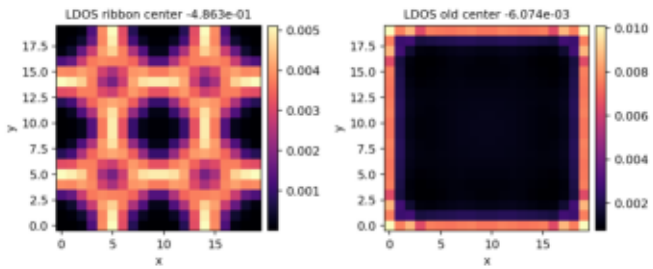
best_edge



rot_edge



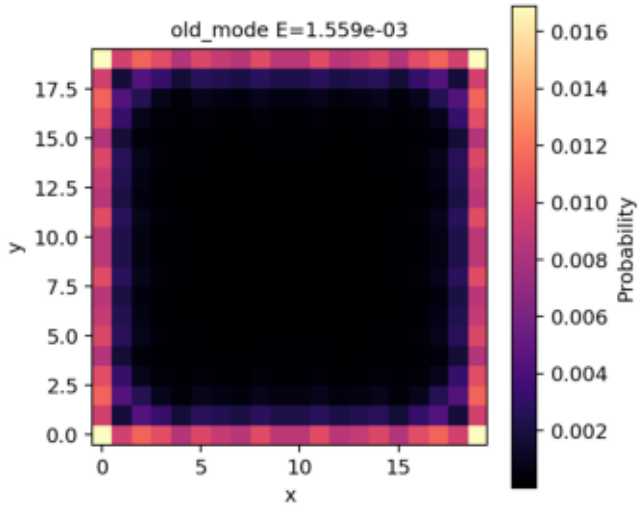
LDOS



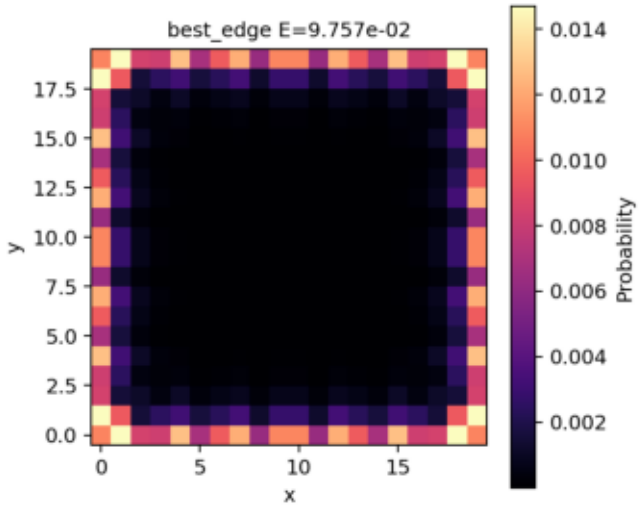
old_edge=0.323, best_edge=0.988, rot_edge=0.987; role=edge_dominated

□□A - v_0p60_t_0p10_lm_0p30
v=0.60, t=0.10, lm=0.30, gap=1.110e-16, Z2=1, Chern=-0.000, Wilson=-1.032

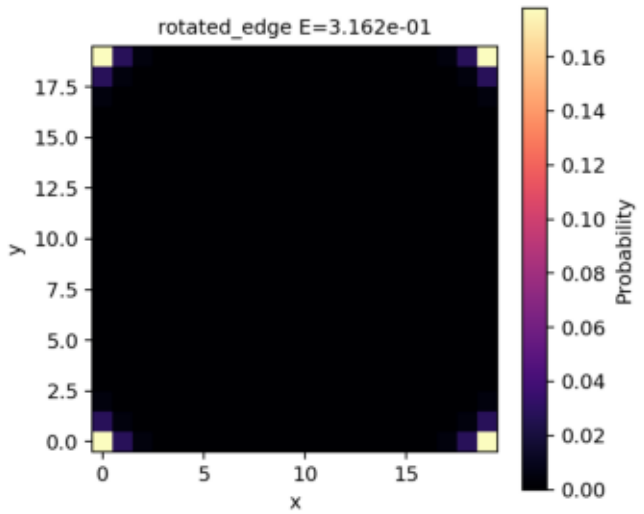
old



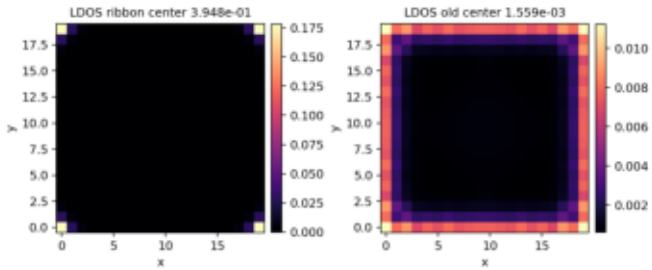
best_edge



rot_edge



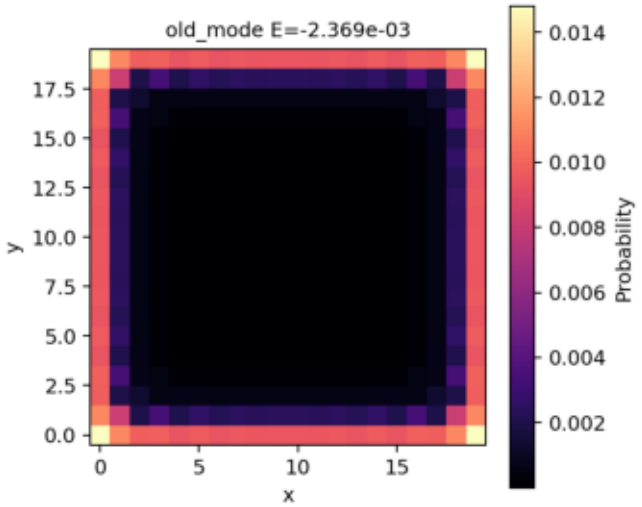
LDOS



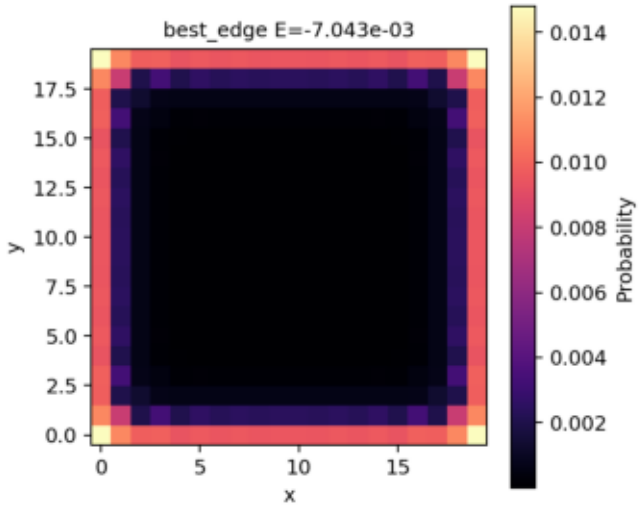
old_edge=0.935, best_edge=0.949, rot_edge=0.999; role=corner_dominated

□□A - v_0p60_t_0p20_lm_0p20
v=0.60, t=0.20, lm=0.20, gap=1.167e-16, Z2=0, Chern=0.000, Wilson=-0.373

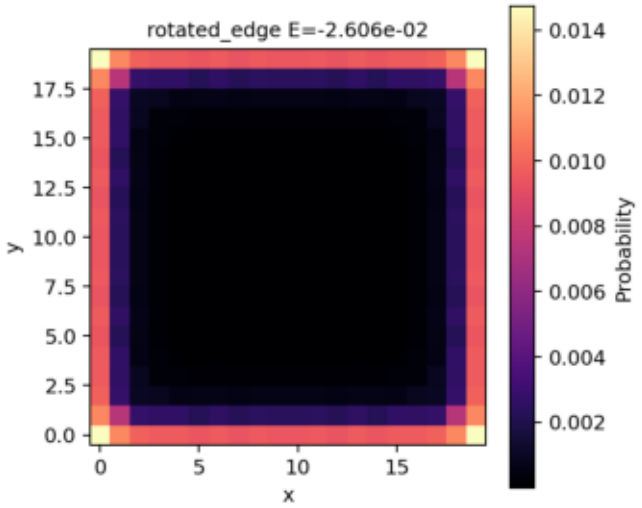
old



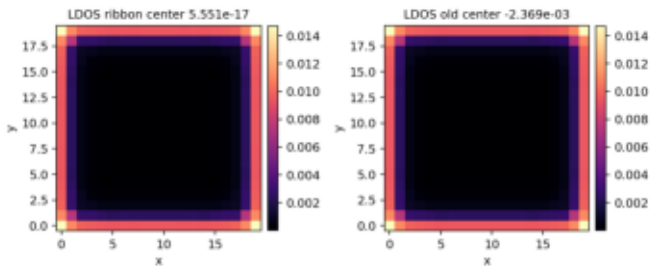
best_edge



rot_edge



LDOS



old_edge=0.949, best_edge=0.949, rot_edge=0.949; role=corner_dominated

□□B □□□□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near-degeneracy □□□□20□□
v_0p60_t_0p90_lm_0p80 dim=2 edge_gain=0.4526 corner_gain=0.1955
v_0p60_t_0p90_lm_0p80 dim=6 edge_gain=0.5353 corner_gain=0.1684
v_0p60_t_0p90_lm_0p80 dim=6 edge_gain=0.4307 corner_gain=0.2013
v_0p60_t_0p90_lm_0p80 dim=6 edge_gain=0.5707 corner_gain=0.1058
v_0p60_t_0p90_lm_0p80 dim=2 edge_gain=0.4074 corner_gain=0.0672
v_0p60_t_0p90_lm_0p80 dim=2 edge_gain=0.4124 corner_gain=0.2537
v_0p60_t_0p90_lm_0p80 dim=6 edge_gain=0.5494 corner_gain=0.1175
v_0p60_t_0p90_lm_0p80 dim=4 edge_gain=0.3232 corner_gain=0.0521
v_0p60_t_0p90_lm_0p80 dim=2 edge_gain=0.3859 corner_gain=0.2109
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v_0p60_t_0p90_lm_0p80 dim=4 edge_gain=0.3427 corner_gain=0.2091
v_0p60_t_0p90_lm_0p80 dim=6 edge_gain=0.5107 corner_gain=0.1373
v_0p60_t_0p90_lm_0p80 dim=2 edge_gain=0.2826 corner_gain=0.1142
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v_0p60_t_0p90_lm_0p80 dim=4 edge_gain=0.2688 corner_gain=0.0797
v_0p60_t_0p90_lm_0p80 dim=2 edge_gain=0.3477 corner_gain=0.0878
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